

2019-2024 SUMMARY REPORT

CHURCHILL RIVER LAKE STURGEON MOVEMENT STUDY



November 2025

Tataskweyak Cree Nation
PO Box 250
Split Lake, MB
R0B 1P0



Dear Chief and Council,

RE: Churchill River Lake Sturgeon Movement Study: 2019-2024 Summary Report

AAE Tech Services Inc. (AAE) is pleased to submit our Technical Report for the Churchill River Lake Sturgeon Movement Study to Tataskweyak Cree Nation. The following report outlines the study area, background information, study methods, results of field investigations from 2019-2024, and outlines the next steps for future studies. Field investigations monitored the movement of 47 adult Lake Sturgeon implanted with acoustic telemetry tags throughout the duration of the open water season during the six study years. Results from this study will assist in determining habitat utilization of Lake Sturgeon in a post-Churchill River Diversion environment marked by drastic changes in flow regimes.

Should you require additional information or have any questions regarding the attached report please feel free to contact myself at 204-997-3483 or via email at mldowdon@aaetechservices.ca.

Sincerely,

A handwritten signature in blue ink, appearing to read 'Mark Lowdon', is positioned above a horizontal line.

Mark Lowdon

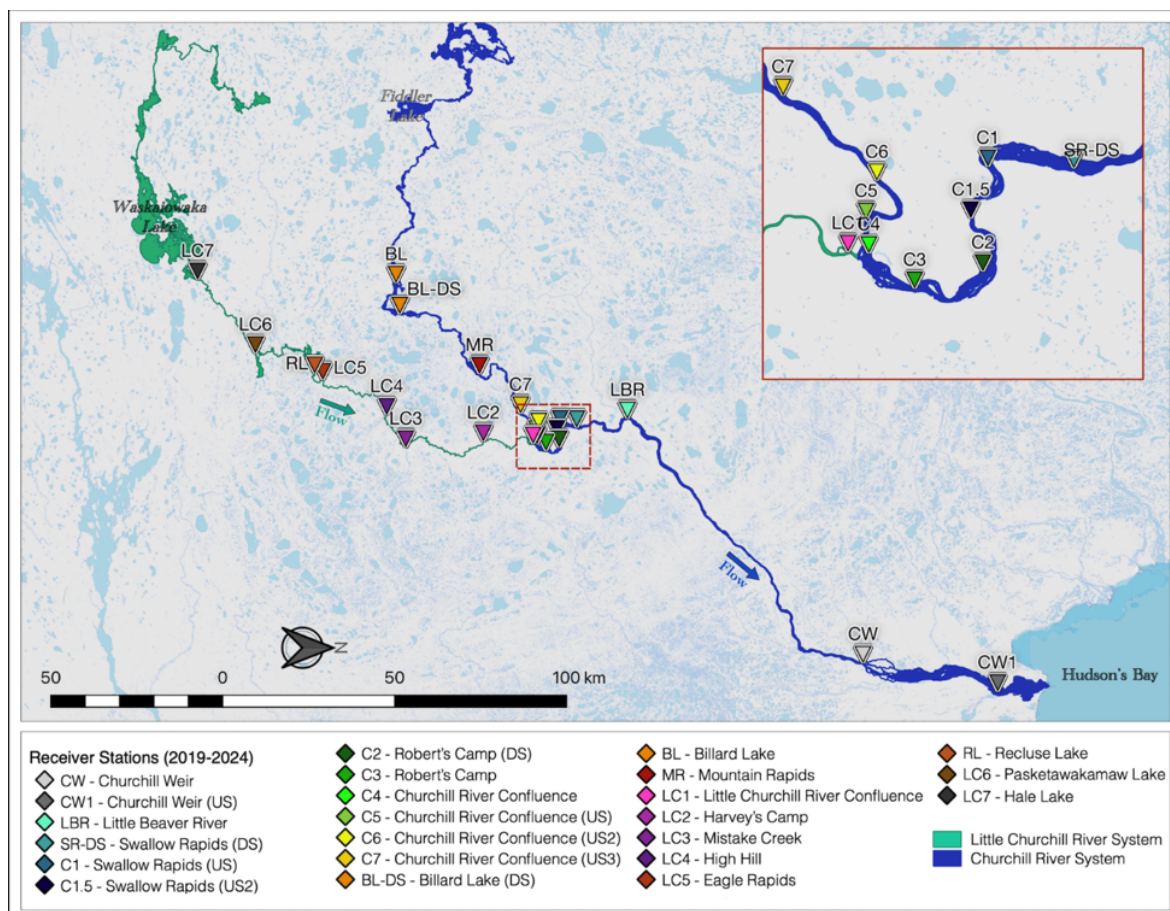
Owner and Project Manager

AAE Tech Services Inc

PROGRAM OBJECTIVES

The objectives for this movement study included four main points:

- (1) Better understand those habitats being selected by Lake Sturgeon for spawning, feeding, refuge and overwintering on the lower Churchill River.
- (2) Determine the extent to which Lake Sturgeon use the Little Churchill River during both the open water season and for overwintering.
- (3) Better understand Lake Sturgeon movement patterns within the Churchill River and Little Churchill River under a hydro regulated river where flow patterns and hydrology are in constant flux.
- (4) Identify barriers to Lake Sturgeon movements during both high and low flow years.



Fish movement is tracked by telemetry receivers deployed at various locations throughout the river system.

WORK OVERVIEW

Lake Sturgeon movement throughout the six-year study period was monitored by acoustic telemetry. In 2019, 47 adult Lake Sturgeon were surgically implanted with acoustic transmitter tags. Each tag emits a unique signal that is detected and recorded by a receiver if a tagged fish enters its detection range (roughly 1 km). This allows each individually tagged sturgeon's movement to be tracked throughout the year. Each spring, acoustic telemetry receivers were deployed throughout the Lower Churchill and Little Churchill rivers to identify, and record, instances of tagged fish at strategic locations. In the Churchill River, receivers were deployed at the confluence of the Churchill and Little Churchill rivers (the Confluence), as far upstream as Billard Lake, and as far downstream as the Churchill Weir. After 2020, locations were focused from just upstream of the confluence to Swallow Rapids. The Little Churchill River locations included the lower reach of the river, roughly 1 km upstream of the Confluence, as well as Harvey's Camp, Mistake River, High Hill River, Eagle Rapids, Recluse Lake, Pasketawakamaw Lake, and Hale Lake. Receiver locations varied slightly from year to year or were adjusted to improve coverage in an area of interest.



Biologist surgically implanting an acoustic transmitter tag inside the abdominal cavity of an adult Lake Sturgeon (2019)

Hourly water temperature (obtained from the acoustic telemetry receivers), and hydrologic data from Water Survey of Canada (WSC) gauging stations were used to assess the influence of temperature, water level and discharge on Lake Sturgeon movement and distribution.

KEY FINDINGS

Flow conditions control connectivity and sturgeon movement.

This study provides strong evidence that changes in water level and discharge directly influence habitat connectivity and, in turn, Lake Sturgeon movement and distribution throughout the Lower Churchill River watershed (including, specifically, the Little Churchill River). When flows are high, sturgeon are able to navigate hydraulic features that are otherwise impassable under low-flow conditions.

High-flow years enabled major dispersal into the Little Churchill River.

During 2020, spring/early-summer 2021, and most of 2022, high flows dramatically increased the number of tagged Lake Sturgeon utilizing habitat within the Little Churchill River, compared to 2019, 2023, and 2024. Water levels at the Churchill–Little Churchill confluence appear to determine whether passage into the Little Churchill River is possible.

The Little Churchill River is heavily used – but only when accessible.

When flows are high, sturgeon dispersed widely throughout the Little Churchill River, traveling between multiple traditionally significant locations and occupying ranges exceeding 135 km, from the Confluence upstream to at least Pasketewakamaw Lake. However, this utilization is seasonally limited. As flows recede in late summer, sturgeon emigrate from the Little Churchill River.

Movement patterns are strongly seasonal and influenced by flow and temperature.

Peaks in movement were consistently recorded in mid-to-late July. Earlier peaks likely occurred in the spring but were missed due to receiver deployment timing. Seasonal water level and temperature changes both influenced these peaks, reinforcing that hydrology is a primary driver of sturgeon movement.

This is the first study linking hydrology and sturgeon movement in the Lower Churchill River.

The findings fill major knowledge gaps about seasonal movement patterns and dispersal limits of this sturgeon population. They demonstrate that when flow conditions allow access, the Little Churchill River is a key seasonal habitat, supporting sturgeon use throughout the summer until declining flows force emigration.

FUTURE DIRECTIONS

Continued study of Lake Sturgeon movement within the Lower Churchill and Little Churchill Rivers is recommended. Increasing the number of tagged individuals and receivers, along with earlier deployment and overwintering receivers can help expand on the conclusions found in this study. Movement studies for juvenile Lake Sturgeon are already underway.

Further investigations into habitat preference – building on the clear relationships already identified between flow conditions, connectivity, and sturgeon use – would refine our understanding of the specific environmental conditions needed to support different life stages. In particular, developing habitat suitability indices (scoring how suitable a site is for sturgeon based on things like flow, depth, and substrate) for both adults and juveniles would provide a structured framework for assessing how hydrology and habitat features combine to sustain, and potentially recover, the sturgeon population in the Churchill River system.

Recovery efforts such as stocking upstream of Mountain Rapids and/or establishing a regulatory minimum flow amount in the Lower Churchill River should be investigated.

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1 General Introduction

1.1 Study Background

Hydroelectric generating stations are a major source of power production in Northern Manitoba, where large rivers such as the Nelson and Churchill drain substantial volumes of water into Hudson's Bay. While these facilities provide renewable energy, they are also associated with a range of environmental effects, including restrictions to fish movement, alterations to flow regimes, increased erosion, and habitat disruption.

Lake Sturgeon (*Acipenser fulvescens*) is a species of fish that are particularly susceptible to the changes brought about by hydroelectric development. They are a long-lived and slow growing fish, reaching sexual maturity between 12-15 years in males and 18-27 years in females (Scott & Crossman, 1973; COSEWIC, 2017). Throughout their lifespan, they can travel great distances to find the habitat required for each of their life stages. While feeding during the warmer months, sturgeon will seek out fine substrates where their prey is more abundant. During the winter, they move to deep, slow-moving pools with higher levels of dissolved oxygen. Finally, during spring spawning, they generally prefer swift current below rapids or beneath waterfalls at the upper reaches of large river systems (Auer, 1996; Holm et al., 2009; Kerr et al., 2011; Stewart and Watkinson, 2004) where there is coarse substrate. Given the varied and specific nature of habitat requirements, it follows that Lake Sturgeon require a large area of suitable habitat which hydroelectric generating stations directly limit.

In the northern region of Manitoba, the Churchill River Diversion (CRD) has arguably had the most profound impact on aquatic ecosystem health and the accompanying Lake Sturgeon populations. The construction of the CRD began in 1973, with the aim of redirecting flow from the Churchill River to the Nelson River, where the additional water would support hydroelectric generating stations on the Nelson system (Manitoba Hydro, 2019). Upon its completion in 1976, approximately 75 – 98.5% of flow (Manitoba Hydro, 2019; Newbury et al., 1984) was diverted from the Churchill River southward into the Burntwood and Nelson Rivers. Construction of the Missi Falls Control Structure (MFCS) at the outlet of Southern Indian Lake effectively segmented the Churchill River in two distinct reaches; the Upper Churchill River, upstream of Missi Falls at Southern Indian Lake, and the Lower Churchill River, below the MFCS and extending to the Hudson Bay.

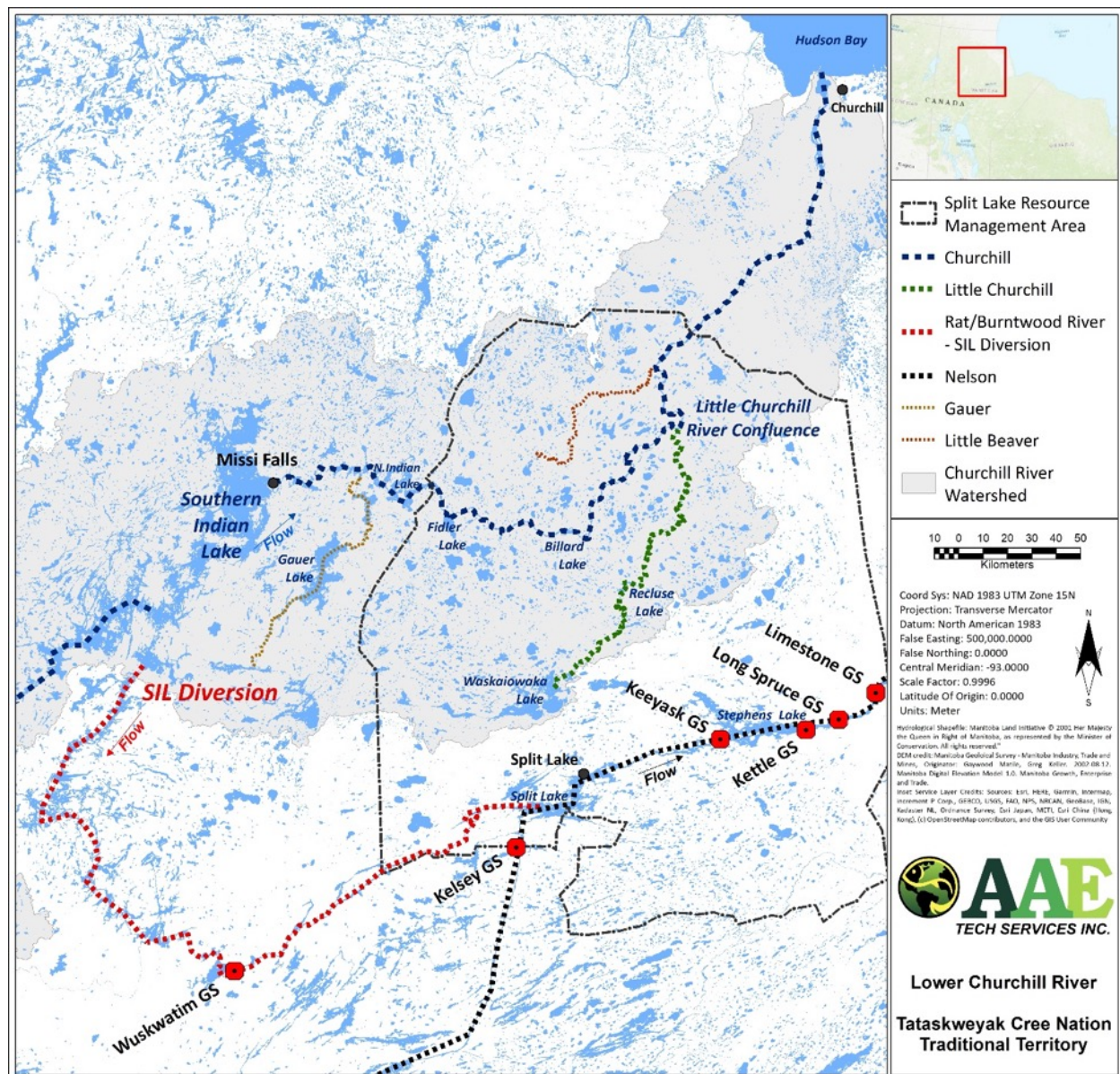


Figure 1. Redirected flow from the Churchill River.

The completion of the CRD led to widespread loss and degradation of fish habitat (Ambrose and McDougall, 2018; Bodaly et al., 1989; Government of Canada; Manitoba Government - Conservation and Water Stewardship Fisheries Branch, 2012; Manitoba Hydro, 2019; Newbury et al., 1984). Lake Sturgeon experienced dramatic reductions in suitable spawning habitat in the Lower Churchill River due to reduced flows and water levels and were barred from accessing historical spawning grounds upstream of the MFCS at South Indian Lake (Ambrose and McDougall, 2018; Bodaly et al., 1989; DFO, 2010; Manitoba Government - Conservation and Water Stewardship Fisheries Branch, 2012).

The CRD not only reduced flows, but also greatly increased the variability of flows and water levels. Flows range from a fraction to approximately double that of the pre-CRD levels which constantly disrupts the environment and prevents the re-establishment of suitable habitat. The impacts of habitat fragmentation and flow variability on Lake Sturgeon success caused by dams are well documented (Auer, 1996; DFO, 2010; Haxton and Findlay, 2008; Haxton and Findlay, 2009; Kerr et al., 2011) and may be the most significant hurdle to population recovery in disturbed environments such as the post-CRD Churchill River (DFO, 2010).

The Churchill River holds major significance to Indigenous Peoples of the North, including the Tataskweyak Cree Nation (TCN), who for centuries relied on an abundant Lake Sturgeon population as an important subsistence fishery and cultural resource. Aboriginal Traditional Knowledge (ATK) held by Elders and resource users from TCN readily recall a not-so-distant past where the area was pristine and sturgeon could be found in large numbers throughout the river.

However, recent decades have seen the river's sturgeon stocks decimated by anthropogenic disturbance (DFO, 2010; Haxton et al., 2015), including overharvest, pollution, and hydroelectric infrastructure development. Since the CRD was built, Lake Sturgeon have been effectively extirpated throughout most of their historic range (Ambrose and McDougall, 2018) all but disappearing from the Upper Churchill River. The remaining population, relegated to only a small portion of the Lower Churchill River, has also seen significant declines with ATK noting that populations have died off sequentially from Northern Indian Lake, to Fiddler Lake, and Billard Lake after that (Pers Comm Robert Spence, 2018). The pattern continued downstream until only one sizeable population remained in the Lower Churchill River at the Confluence of the Little Churchill River.

In 2017, the Saskatchewan – Nelson River and Western Hudson Bay populations of Lake Sturgeon were classified as endangered by the *Committee on the Status of Endangered Wildlife in Canada* (COSEWIC) and are currently being considered for listing under Schedule 1 of the federal Species at Risk Act (SARA) (Government of Canada, accessed 2021).

1.2 Scope of Work

AAE Tech Services Inc. was contracted by Tataskweyak Cree Nation (TCN) to perform assessments of Lake Sturgeon distribution, habitat use, and movement patterns within the Churchill and Little Churchill rivers over the years of 2019-2024. For every year of the study, the integration and use of ATK provided by TCN was paramount in the analysis of pre- and post-CRD development distribution of sturgeon throughout the Churchill River watershed

and in identifying historically significant sturgeon habitat throughout the Little Churchill River. The value of integrating traditional knowledge with scientific research is widely recognized in environmental governance and resource management (Berkes et al., 2000; Tengö et al., 2014).

How Lake Sturgeon are utilizing the remaining habitat after disruption from the CRD was one of the main objectives of this study. This includes a close examination of the extent to which sturgeon are using the Little Churchill River (LCR) and whether they use the LCR for spring spawning.

Acoustic tags were used to follow Lake Sturgeon movement during the open water seasons and at select locations during the winter months. Areas where sturgeon would congregate were assessed to better understand the type of local habitat that is being utilized during the fulfillment of different life stages of the sturgeon. Additionally, sturgeon movement between areas was compared with various environmental parameters including flow rates, water depth, habitat/food availability, water temperature, and season, in an effort to understand if and how these parameters affect their movement. This study focused on adult Lake Sturgeon (>800mm total length) with future studies on juveniles forthcoming.

1.3 Sturgeon Background

The life history traits of Lake Sturgeon (*Acipenser fulvescens*) are typical of the fishes within the family Acipenseridae. These are long natural lifespans (up to 154 years for females), low growth rates, and very large adult sizes. Their reproductive life history is unusually slow, characterized by protracted generational times as a consequence of late-onset maturity (12-15 years in males, 18-27 years in females) and low reproductive rates compared to other species that occupy similar ranges and ecological guilds (Ambrose and McDougall, 2018; Auer, 1996; Bruch et al., 2016; Holm et al., 2009; Kerr et al., 2011; MacKay, 1963; Peterson et al., 2007; Stewart and Watkinson, 2004).

The diet of Lake Sturgeon is almost entirely benthic in origin and is predominantly comprised of aquatic insects, crayfish and other benthic (bottom-dwelling) macroinvertebrates (Kerr et al., 2011; Stewart and Watkinson, 2004). Adult sturgeon seek out fine substrates comprised mainly of sand, silt, and clay as they tend to be associated with higher densities of these prey items (Kerr et al., 2011). They may shift from riverine habitats to floodplains or estuarine areas, where river and lake environments mix, depending on prey availability.

Spawning occurs in the spring, from mid-April to early June, when water temperatures reach 10-15°C (Holm et al., 2009; Kerr et al., 2011; Peterson et al., 2007; Stewart and Watkinson, 2004). Sturgeon typically migrate upstream into the accessible upper reaches of large river systems where they spawn in swift currents over gravel or cobble, often at the base of impassable barriers such as waterfalls, rapids or dams (Auer, 1996; Kerr et al., 2011; Peterson et al., 2007; Stewart and Watkinson, 2004). The availability of suitable spawning habitat is thought to be of great importance, and it has been estimated that a single female sturgeon requires approximately 13- 48 m² of habitat in order to maximize spawn success rates (Fortin et al., 2002).

Overwintering habitat for Lake Sturgeon are usually areas of reduced flow and high oxygen content. Sturgeon will reside in these areas, often deep sections of rivers or adjoining lakes, for the winter months where they will typically exhibit little movement and conserve energy. This allows the large fish to survive the winter when metabolic rates are low and food is scarce. The yearly migration of sturgeon in their search for food, spawning grounds, and overwintering sites can cover significant distances of tens to hundreds of kilometers.

2 Methods

2.1 Physical and Hydrologic Monitoring

The hydrological characteristics of the river were recorded to contextualize sturgeon movements within the system over years of highly variable conditions. Daily water level and discharge¹ data were retrieved from the Water Survey of Canada gauging station records (Government of Canada) for the Churchill River below Fidler Lake gauging station (06FB001), and the Little Churchill River below Recluse Lake gauging station (06FC001) for all study years. Hourly temperature data was retrieved from the Vemco VR2Tx Acoustic Receivers deployed within the Churchill and Little Churchill Rivers.

2.2 Fish Movement Tracking

Throughout the study period, movement of tagged sturgeon was monitored by using Vemco 69 kHz V16 coded internal transmitter tags. In 2019, 47 adult Lake Sturgeon were surgically

¹ In this report, “discharge” refers to the measured volume of water (m³/s) at gauging stations, while “flow” is used more generally to describe relative river conditions (e.g., high vs. low flow years). “Water flows” is used interchangeably with “flow conditions.”

implanted with acoustic transmitter tags. Each tag emits a unique signal that is detected and recorded by a receiver if a tagged fish enters its detection range (roughly 1 km, varying based on depth, water quality and substrate). This allows each individually tagged sturgeon's movement to be tracked throughout the year. Between 10-11 Vemco receivers were deployed each year, with some variation in site choices between years based on accessibility and improvements in knowledge from year to year. In 2019 and 2020, receivers were deployed across upstream and downstream reaches of the Lower Churchill to document broad-scale distribution, but in subsequent years, deployment was concentrated at the Confluence and Little Churchill River system, where initial results and ATK indicated the highest sturgeon activity. Selected sites included multiple locations around the confluence of the Churchill and Little Churchill River, and the stretch of the Little Churchill River (LCR) from Recluse Lake to the Confluence. Sites as far downstream as the Churchill River Weir, and as far upstream as Hale Lake and Billard Lake, were investigated. Locations were identified in direct consultation with local TCN members.

During surgical preparations, sturgeon were kept in a shallow section of the river in proximity to the surgery station and held by tail ropes anchored to the shore (Figure 2) to reduce confinement and handling stress prior to surgery. Surgical procedures followed protocols developed by DFO and were similar to those described by Haxton (2003). An employee from DFO was present to supervise the procedure.



Figure 2. TCN participant restrains sturgeon near shore. Ropes are attached to tails so that they may be kept in the water prior to surgery.

Surgical instruments were disinfected and sterilized using a diluted betadine solution. Fish were then brought to the surgery table and manually restrained, ventral side facing up and their gills were aerated and kept wet using a 1,000 ml Nalgene wash bottle (MilliporeSigma Canada, Oakville, ON, CAN) filled with river water throughout the duration of surgery. The incision site on the abdomen was disinfected with the same solution used to sterilize surgical instruments. A small, 2-3 cm long, shallow incision was

made adjacent to the midline of the abdomen, below the ribs (Figure 3A). Tags were then inserted through the incision with a posterior angle and direction until they were no longer

visible at the point of incision (Figure 3B). Incisions were sealed using dissolving suture thread (Figure 4) and washed with betadine solution before fish were returned to the water and held upright until they were able to swim away.



Figure 3. Surgical insertion of Vemco acoustic transmitter tag. (A) A small incision is made lateral to the midline of the abdomen to allow for the insertion of the tag. An assistant aerates the fish's gills with fresh river water throughout the surgery. (B) A Vemco 69 kHz V16 coded transmitter tag is inserted into the abdominal cavity of the sturgeon through the incision.

The morphological data and identification of the individual adult sturgeon that were tagged for this study can be found in the appendix (Table A1).

The signals from the Vemco transmitters were detected by Vemco VR2Tx receivers deployed at various locations in the Churchill and Little Churchill Rivers. The exact locations can be seen in Figures 5 and 6, with the co-ordinates and timing of the Vemco receivers found in Tables A2 – A7 in the appendix.



Figure 4. Dissolving sutures applied to the incision.

Receivers were programmed using Vemco's VUE software (version 2.6.1) to initialize recording and assign UTC offsets to ensure the accuracy of timestamps in detection records. Receivers were fastened to rods using 150 lb-capacity zip-ties and anchored to large stones

(Appendix B; Fig. B1) tethered together by a retrieval line with individual receivers assigned to and deployed at each location in the center of the channel to maximize signal coverage across the width of the river. Receivers in the Little Churchill River and Lower Churchill River were generally deployed in late-June/early-July and were retrieved in the Fall (late-September/early-October). Two receivers were left in the Churchill River over the winter from 2023 to 2024 at locations C3 – Robert’s Camp and C5 – the Confluence, as a preliminary look into the sturgeon’s overwintering behavior. Data was offloaded into Vemco’s Fathom Central software for analysis. Receiver locations varied year to year, based on flow conditions (i.e. safe locations to set and retrieve receivers), research objectives, and the desire to improve the signal coverage in a particular area based on previous years data.

The receivers in the Churchill River in 2020 (Billard Lake, Mountain Rapids, the Confluence, Little Beaver River, and Swallow Rapids) that were unable to be recovered in the fall of 2020 due to high flows remain in the river at the time of writing and are presumed to have recorded data until their batteries ran out approximately 18 months after they were activated. It is unlikely they will be found and are considered lost.

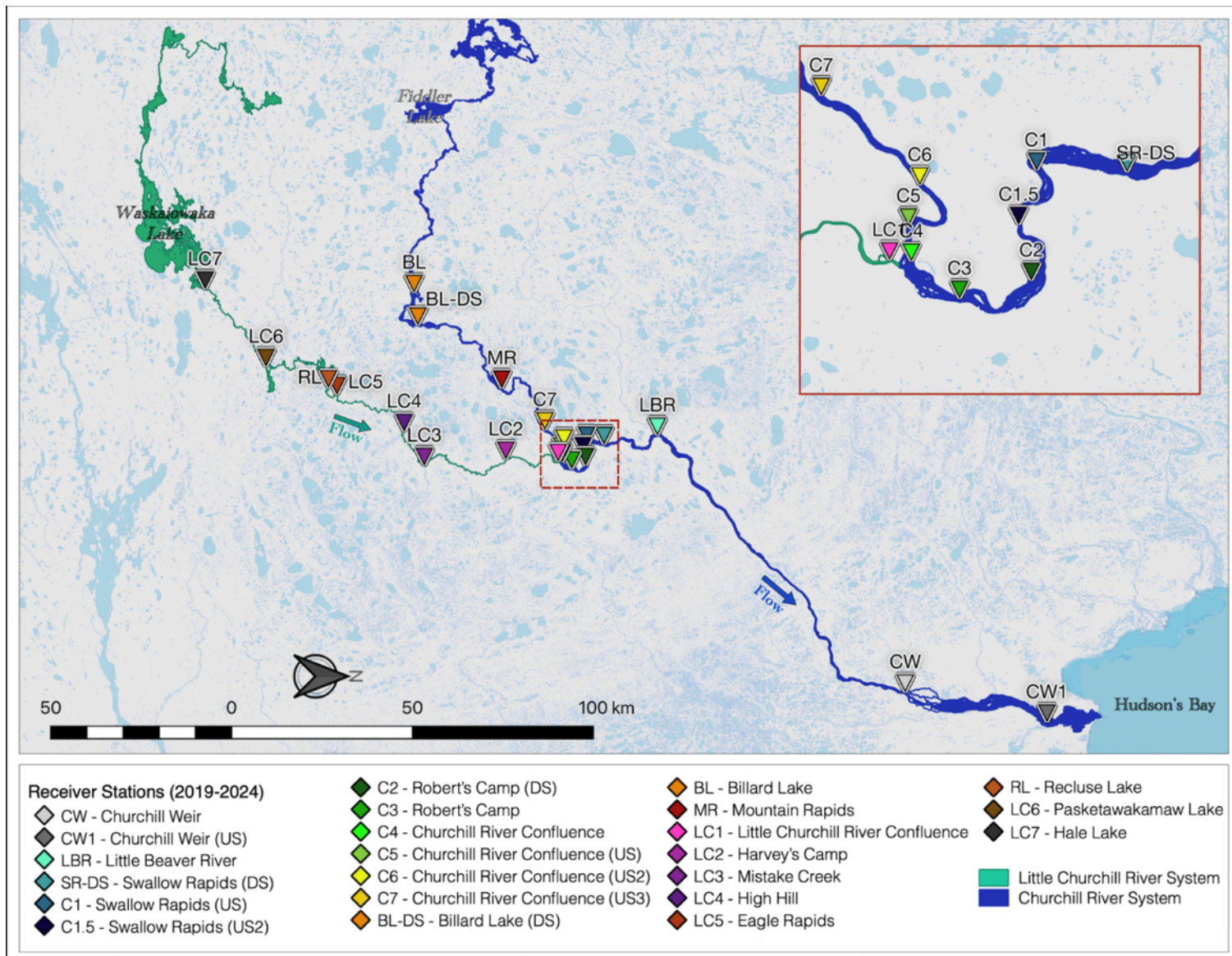


Figure 5. Vemco Receiver locations for all study years.

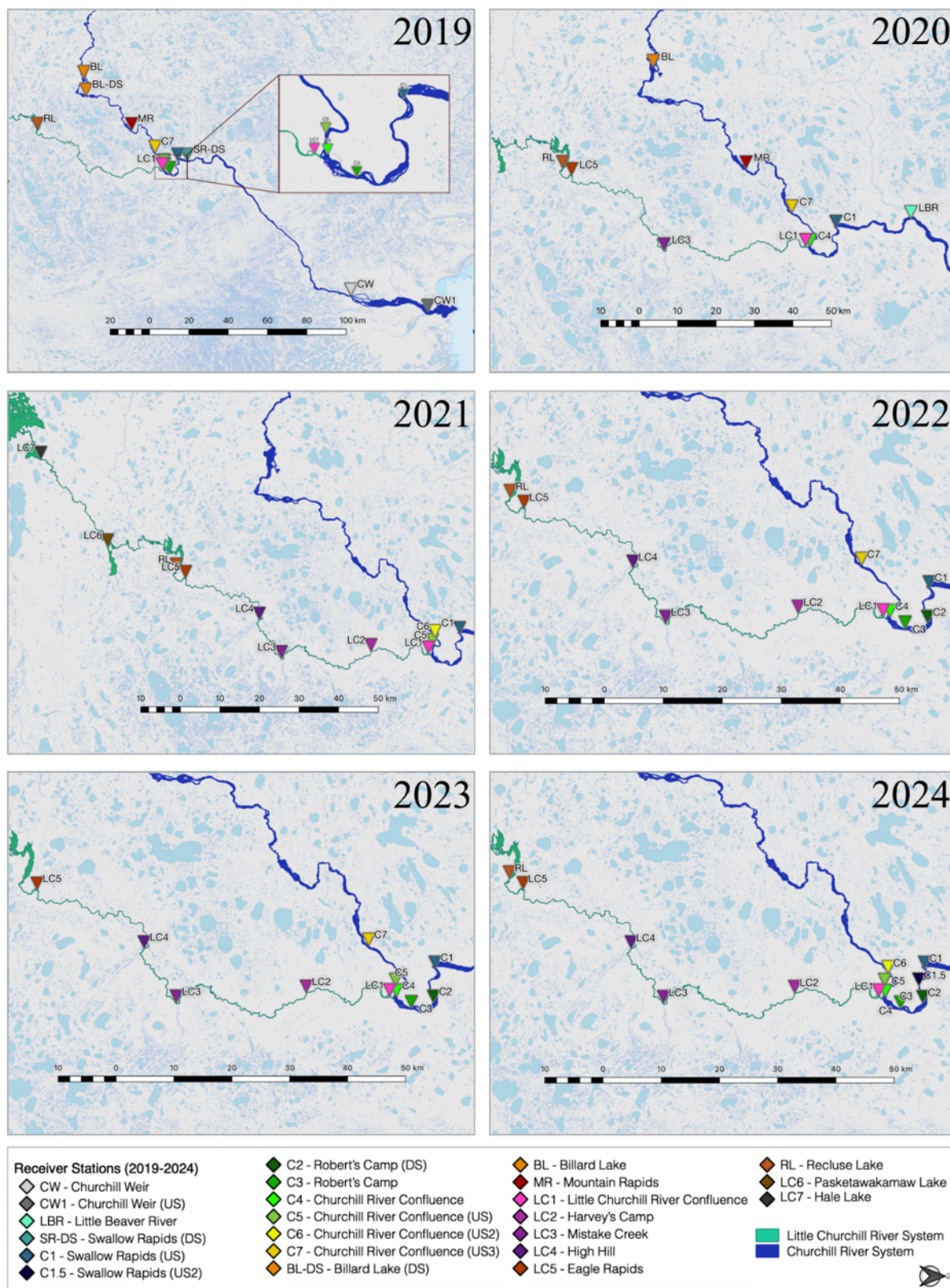


Figure 6. Vemco receiver locations for each individual year.

2.3 Sonar Passage Estimates (2020)

A Sound Metrics Dual-Frequency Identification Sonar (Didson; Sound Metrics, Bellevue, WA, USA) was installed in the Little Churchill River to evaluate passage of adult sturgeon into the river that may be associated with spawning migrations during the 2020 study year. The sonar was installed at the downstream extent of an island located approximately 1 km upstream of the confluence with the Churchill River, immediately upstream from the LC1 (Little Churchill/lower-LCR) telemetry receiver (see Figure 3).

Owing to its deeper, narrower channel and lesser flow velocity, the channel along the west side of the island was selected as the primary location for the sonar (see Appendix B; Fig. B4) and was expected to be the preferred movement corridor taken by fish swimming upstream from the Confluence. From the north (downstream) point of the island, the sonar was installed facing the western shoreline, perpendicular to flow. The Didson sonar and its mounting apparatus were secured to pickets anchored in the riverbed and on shore, leaving the sonar in the water to operate continuously without requiring the presence of an operator (see Appendix B; Fig B5). A weir constructed using pickets and snow fencing was angled out from shore and extended out front of the sonar beam. This served to direct fish moving upstream into the visible range of the sonar (see Appendix B; Fig. B5).

The weir was inspected daily to remove accumulated debris, ensure no fish were becoming trapped, and to verify that the integrity of the weir was maintained to prevent fish from swimming underneath, over, or around it.

Sonar imaging and recording parameters were programmed using Sound Metrics Didson V.5.26.26 software. Due to high-water levels and the width of the Little Churchill River within the reach proximal to the Churchill River, only “low-frequency” (LF) imaging was used to maximize sonar coverage across the river channel. Recording parameters were programmed to automatically store recorded piSonar files hourly, with recording initiated at approximately 11:00 on June 12th, continuing until 12:18 on June 19th, 2020. On June 19th, the sonar and weir were removed and installed facing into the east channel, to evaluate potential passage on this side of the island, despite being generally shallower and exhibiting more rapid and turbulent flow. Once in place, recording was re-initialized at approximately 13:50 on June 19th and continued until approximately 23:04 on June 21st, 2020.

2.4 Analysis Methods

Collecting transmission data for all tagged sturgeon over six years resulted in millions of receiver pings. Analysing these data with consideration for the hydrological and chronological data represents a challenge. As a first step, repeated pings for individual fish were reduced to hourly detections. The Vemco tags implanted into the sturgeon, on average, send a signal every few minutes. For every hour that a tagged fish stays in the vicinity of a receiver, this operation reduced the computational burden by about 30 times.

After reducing the detections to an hourly basis, the data was analysed for potential false detections. Any fish that were detected at distant locations relative to their last known location were removed. The rough cut-off for this is when detections would require prolonged swimming speeds of over 10km/hr. Given that Lake Sturgeon have a two-minute critical swimming speed of about 1.5km/hr (Webb, 1986), this higher cut-off leaves room to account for flow speeds and the fact that distances were calculated on the GPS co-ordinates of receivers while detections can occur at the edges of their detection radius. This should remove definite false positives while leaving the majority of true data unaltered.

After the data was reduced, it was examined for general patterns in movement. Initially, all individual fish movements were plotted on a time axis with colour coded receiver locations, with each year as a separate figure (Figures A7-A12 in Appendix A2). After examining the data, several individuals were selected for closer illustration: some because their movement patterns were unusual, and others because they typified the broader patterns observed across the population. These case examples were used to highlight key findings. The locations and timings of the general movement were based off of these initial analyses.

To gain a deeper understanding of the Lake Sturgeon's movements throughout the system, the movement data was compared to both temperature and flows. First, a distance matrix for the river length between receiver locations was created. The river kilometers between the GPS co-ordinates of the receiver locations was taken to be the distance between receivers. Then, both the distances travelled by individual fish and the total distance travelled by all fish were plotted on a weekly basis and reported in yearly heat maps. The average flow was compared to the movement totals for corresponding weeks and plotted to assess for potential associations. Additional heat maps were created for movement within the LCR where upstream movements were considered positive distances and downstream movements negative distances to investigate what factors may play a role in the movement in and out of the LCR.

3 Results

Lake Sturgeon movement within the Lower Churchill and Little Churchill Rivers (LCR) was shaped by three main factors. First, flow levels played a decisive role: high flows consistently allowed sturgeon to access and remain in the LCR, while low flows restricted access and shortened their use of this habitat. Second, seasonality strongly influenced movement, with sturgeon entering the LCR in June, reaching peak abundance in July, and beginning to emigrate in August. Finally, water temperature shifts affected both the timing and intensity of movement across different flow conditions.

The following subsections present the detailed results that underpin these patterns.

3.1 Tagging and Receiver Deployment

The V16 internal transmitter tags together with the Vemco VR2Tx acoustic receiver array proved to be highly effective at monitoring the movement of the tagged sturgeon over the past six years of study. A cumulative total of 59 receivers were set and retrieved for a sum of over 150,000 hours and over 1,000,000 detections of the tagged Lake Sturgeon. At the end of 2024, 44 of the original 47 fish were still being detected in the receiver array. The overall characteristics of the tagged fish can be found in the two tables below, with all details for each fish available in Table A1 in the appendix.

Table 1. Counts of qualitative characteristics.

Sturgeon Tagged	Mature	Immature	Unknown	Male	Female	Unknown
47	20	26	1	11	7	29

Table 2. Quantitative qualities of tagged Lake Sturgeon.

Total Length (mm) Average	Total Length (mm) Range	Weight (kg) Average	Weight (kg) Range	Age Average	Age Range
1217	928 - 1562	11.45	1.7 - 27.9	28.8	15 - 48

The Vemco tag implantation appeared successful for all but one fish. Initial detections began on June 10th, 2019, at the C2 receiver at the Confluence (Figure 20), where surgical implantation of transmitter tags occurred. Most fish remained within detection range of the C2 location for the first two weeks, with some moving nearby to the C1 location. Larger movements were detected after two weeks, presumably after the majority of fish had recovered from the surgery.

The only casualty is sturgeon #7, which was accidentally killed in a small mesh-net used by TCN members in Recluse Lake. The net was not intended for Lake Sturgeon.

Throughout the study, the location of the Vemco receivers shifted from the Churchill River to the Little Churchill River. The main driver of this refocused deployment came after 2020, when it became clear that most of the tagged fish were occupying the LCR whenever flow conditions permitted them to do so, suggesting that low flows may restrict them from reaching this preferred summer habitat. The following table summarizes the location sets and timing of the Vemco receivers for each of the study years.

Table 3. Yearly Vemco receiver set summary.

Year	Number of Receivers Set and Retrieved	Location, Codes	Average Deployment Time (hours)
2019	12	MR, SR-DS, BL, BL-DS, C1, C3, C4, C5, CW, CW1, LC1, LC2	2390
2020	4*	LC1, LC3, LC5, RL	2232
2021	11	C1, C5, C6, LC1, LC2, LC3, LC4, LC5, LC6, LC7, RL	2455
2022	11	C1, C2, C3, C4, C7, LC1, LC2, LC3, LC4, LC5, RL	2398
2023	10	C1, C2, C3, C4, C7, LC1, LC2, LC3, LC4, LC5	2532
2023 Overwinter	2	C3, C5	9036
2024	10	C1, C1.5, C2, C6, LC1, LC2, LC3, LC4, LC5, RL	3355

Location Codes

MR – Mountain Rapids; SR-DS – Swallow Rapids Downstream; BL – Billard Lake; BL-DS – Billard Lake Downstream; C1 – Swallow Rapids; C2 – Robert's Camp; C3 – Robert's Camp Downstream; C4 – Churchill River Confluence; C5 – Churchill River Confluence Upstream; C6 – Churchill River Confluence Upstream 2; C7 – Churchill River Confluence Upstream 3; CW – Churchill Weir; CW1 – Churchill Weir Upstream; LC1 – Little Churchill Confluence; LC2 – Harvey's Camp; LC3 – Mistake Creek; LC4 – High Hill; LC5 – Eagle Rapids; LC6 – Pasketawakamaw Lake; LC7 – Hale Lake; RL – Recluse Lake.

*A total of nine receivers were deployed this year but due to unsafe conditions from high flows, only four were retrievable.

Each unique Vemco receiver code, the GPS co-ordinates of locations, and the individual deployment and retrieval times, can be found in Tables A2 – A7.

3.2 Individual Movement Patterns

While the receivers generated millions of signal detections, it is possible to pull out individual fish movements to help understand what the general movement patterns are, as well as any unique movements that may highlight important study locations. A random subset of representative individuals was selected for presentation. The location data for several representative fish are presented below in Figures 7 – 10.

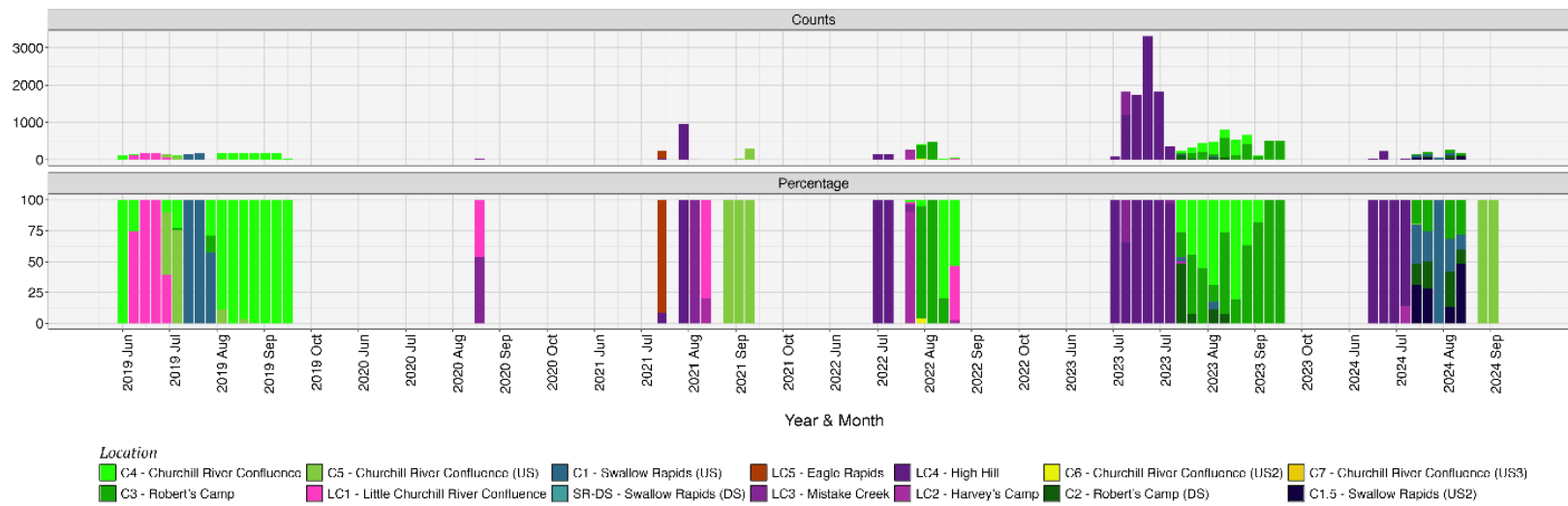


Figure 7. Weekly movements of fish #2. Top panel represents number of detections at each receiver, and bottom panel represents the percentage of detections at each receiver for a given week.

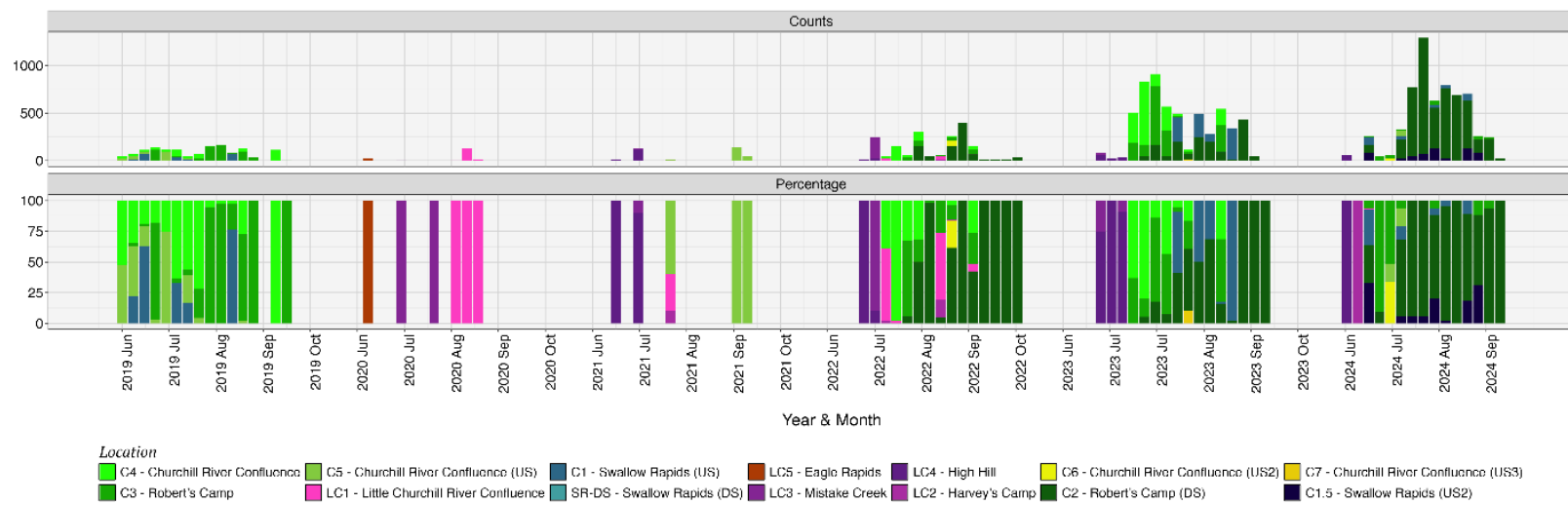


Figure 8. Weekly movements of fish #37. Top panel represents number of detections at each receiver, and bottom panel represents the percentage of detections at each receiver for a given week

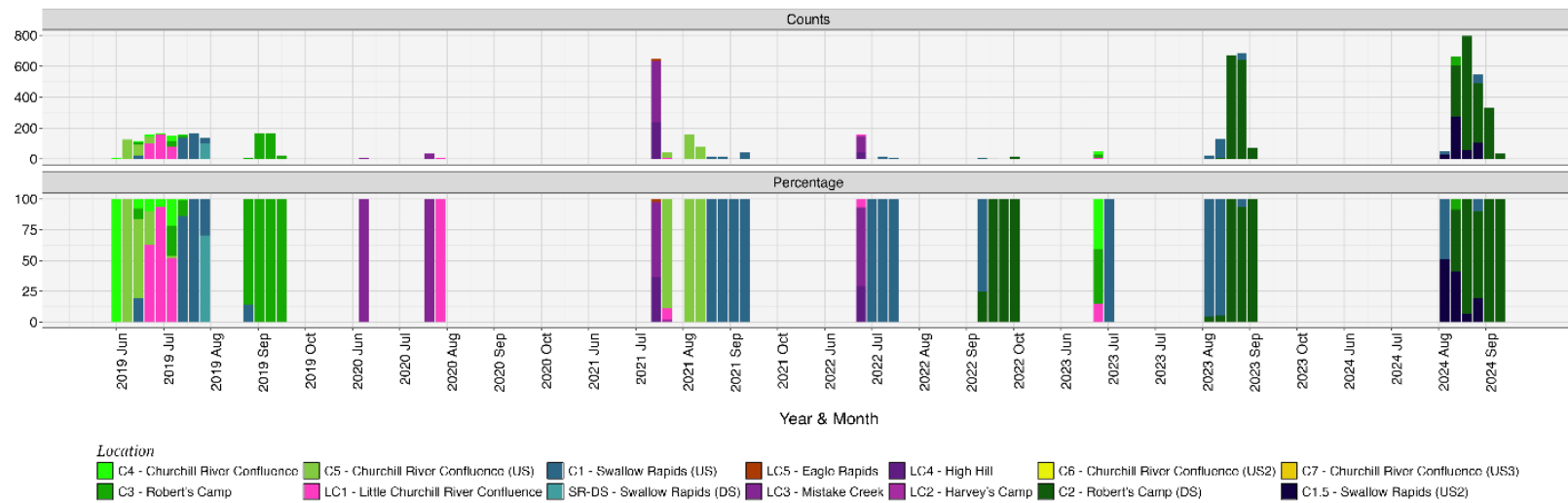


Figure 9. Weekly movements of fish #43. Top panel represents number of detections at each receiver, and bottom panel represents the percentage of detections at each receiver for a given week.

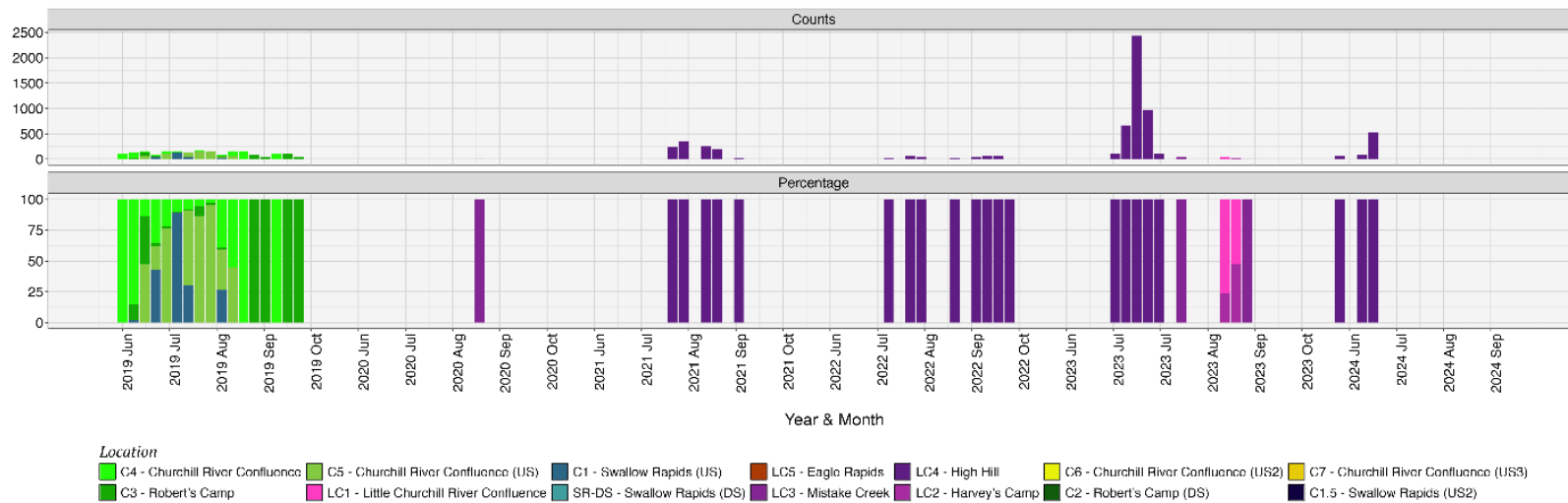


Figure 10. Weekly movements of fish #13. Top panel represents number of detections at each receiver, and bottom panel represents the percentage of detections at each receiver for a given week.

Figure 7 follows the movements of fish #2. Each year, the fish was detected in the Little Churchill River (LCR) (purple/pink) in the spring/early summer and then would return to the Confluence (greens). A similar pattern is seen in Figures 8 and 9 which follow fishes #37 and #43 respectively, with the exception that fish #43 does not make it to the LCR in the lower flow years of 2023 and 2024. Fish #13 (movements shown in Figure 10) was detected through all open months in the LCR from 2020 – 2024, suggesting that the fish did not return to the Confluence for the winter, or perhaps returned much later than others, after the receivers were removed. The majority of detections for #13 were at the High Hill receiver, where a large number of other sturgeon were detected. It is likely there is either favourable habitat or a barrier to further upstream movement at that location (See Figures A13 – A18 in Appendix A2 for proportions of detections at each location).

There were a few other noteworthy individual sturgeon in their use of the LCR. Two individuals, #3 and #26, were detected as moving from the Confluence to Recluse Lake every year that receivers were set there. When moving from the Confluence to get there, it represents a travel distance of over 100 km. It is possible the sturgeon are travelling this great distance to reach desirable habitat either for spawning or feeding.

3.3 Seasonal Use of the LCR

When considering these individuals and the movement data as a whole, there were several large-scale movement patterns that were repeated over the six study years. Generally, when receivers were deployed in late June or early July, many of the tagged fish were already occupying the LCR (see Table 7 for percentages) indicating that sturgeon were moving upstream into the LCR sometime in June. During 2020, Didson sonar records were collected at the mouth of the LCR to confirm the timing of the upstream movement. During the operational period of June 12th to 21st (including records from both the east and west channels), a total of 101 adult Lake Sturgeon (≥ 1 m) were identified in sonar records (based on criteria described in Appendix A4). These included observations of 83 counts with upstream trajectories (+83) and 18 counts with downstream trajectories (-18) (Figure A26).

Lake Sturgeon most heavily used the LCR in July, and they would start to leave as a group by August as indicated by lower percentages in Table 7 compared to July. By September, nearly all Lake Sturgeon have left the LCR with less than 10% of tagged individuals remaining during this month. Most fish spend this time near the Confluence. For the two years that receivers collected data into October, there were no detections of sturgeon in the LCR during that month.

3.4 Influence of Flow Conditions

The length of this study allowed enough time to capture different yearly flow conditions in the river system. There were low flow conditions in 2019, 2023, and 2024, and there were high flow conditions in 2020, 2021, and 2022, with extremely high flows in 2020 (see Figure 14). Capturing data across these different flow conditions highlighted different usage of the LCR. The biggest difference was seen in August, where in low flow years only 2.1 – 22.7% of the originally tagged fish can be found in the LCR, whereas in higher flow years, anywhere from about 31.8 to 76.1% can be found at some point in the LCR. The highest flow year, 2020, corresponds to the longest usage of the LCR, although temperature may have also played a role in this finding.

Table 4. Percentage of tagged fish detected in the LCR by month each year.

Year	June	July	Aug	Sept	Oct	Total
2019	8.5%	8.5%	2.1%	6.4%	N/A	14.9%
2020	47.8%	47.8%	76.1%	N/A	N/A	93.5%
2021	39.1%	93.5%	41.3%	2.2%	N/A	95.6%
2022	N/A	68.2%	31.8%	9.1%	N/A	77.3%
2023	N/A	34.1%	22.7%	2.3%	0.0%	50.0%
2024	27.3%	54.5%	6.8%	0.0%	0.0%	63.4%

Any detections at LC1 or further upstream the LCR at any time during the month is counted towards the total.

The reduced number of Lake Sturgeon in June and July of 2019 can be partially explained by two factors. One is that it was the lowest flow in the LCR recorded during this study, very likely restricting access to the river. The other is that it was the year the fish were tagged. It was expected that the sturgeon would exhibit reduced movement while recovering from the tag implantation surgery which appeared to be the case in the first month of June, especially during the first two weeks.

Extremely high flows in the Churchill River and LCR characterized the 2020 study year. While the additional water likely gave Lake Sturgeon easier access to a larger portion of the river system, the abundance of water caused dangerous conditions and it was possible to retrieve only four of the nine set receivers that year, all from the LCR. This resulted in no coverage of the Churchill River area and the fewest detections by year.

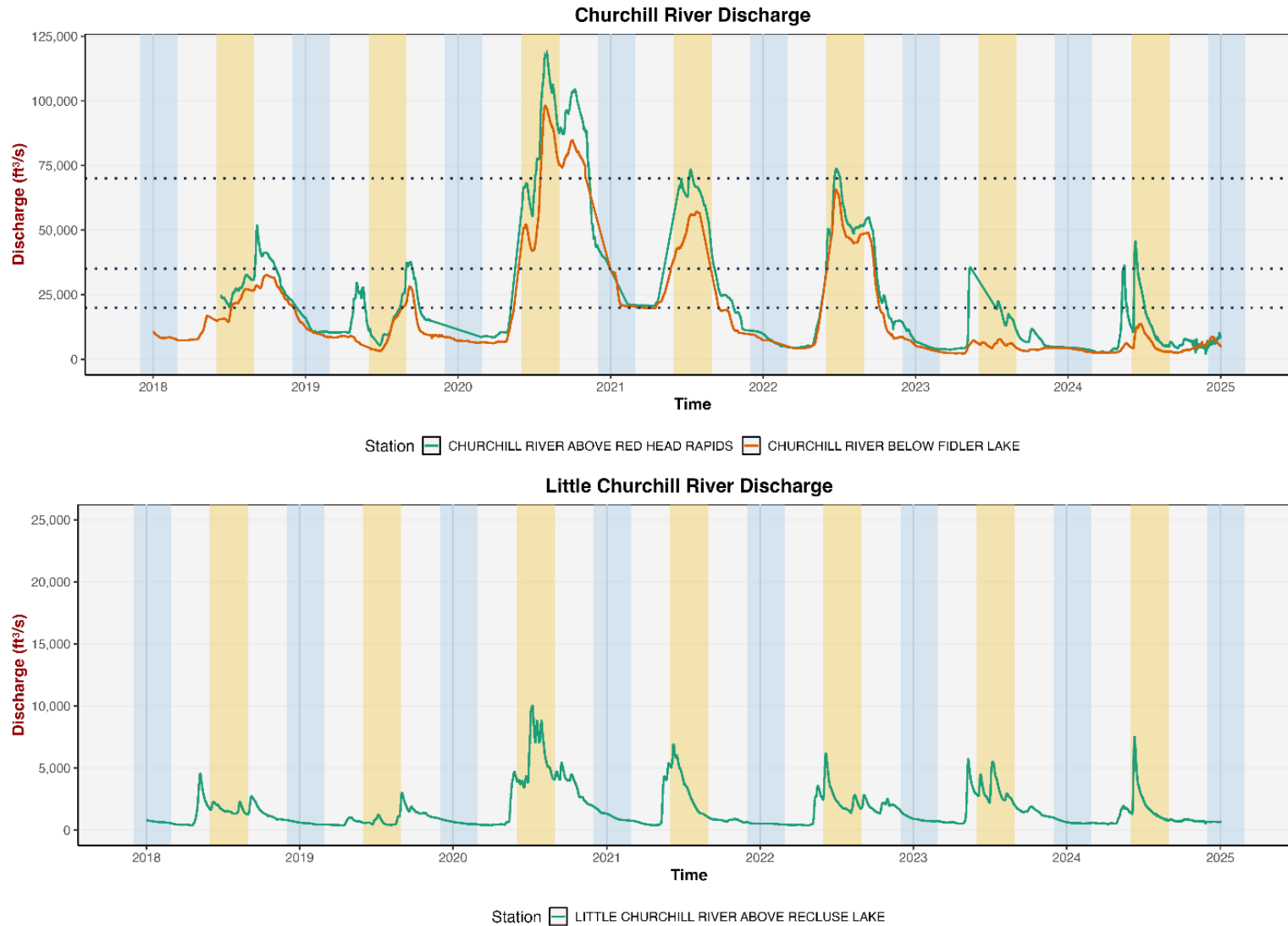


Figure 11. Discharge from the Churchill and Little Churchill Rivers from 2019-2024. The dotted lines at 20 000, 28 000, and 72 000 represent the historical low, average, and high discharge levels from before the construction of the Churchill River Diversion. Summer and winter months are indicated by yellow and blue colours, respectively.

Of the fish that do not go to the LCR, they are typically found within the Churchill River near the Confluence. The data from 2019 gives insight into this movement as this is the year that the receivers were spread mainly through the Churchill River. Most sturgeon stayed within the Churchill River between the two main barriers to the study area, Swallow Rapids downstream, and Mountain Rapids upstream. A few fish did appear to move past these barriers and later return to the study area. In the later years where more receivers were focused on the LCR, areas as far upstream as Mountain Rapids were not covered. However, other receivers placed upstream (locations C6 and C7) show limited movements past the study area in that direction.

The flow rates impact not only the amount of sturgeon travelling into the LCR, but also the movement within the area of the Confluence. During high flow years, more sturgeon are travelling into the LCR as discussed, but fewer sturgeon are being detected at Swallow Rapids, downstream of the Confluence. The inverse is true for low flow years, fewer sturgeon make it into the LCR, but more are detected at Swallow Rapids. Figure 12 depicts the effect.

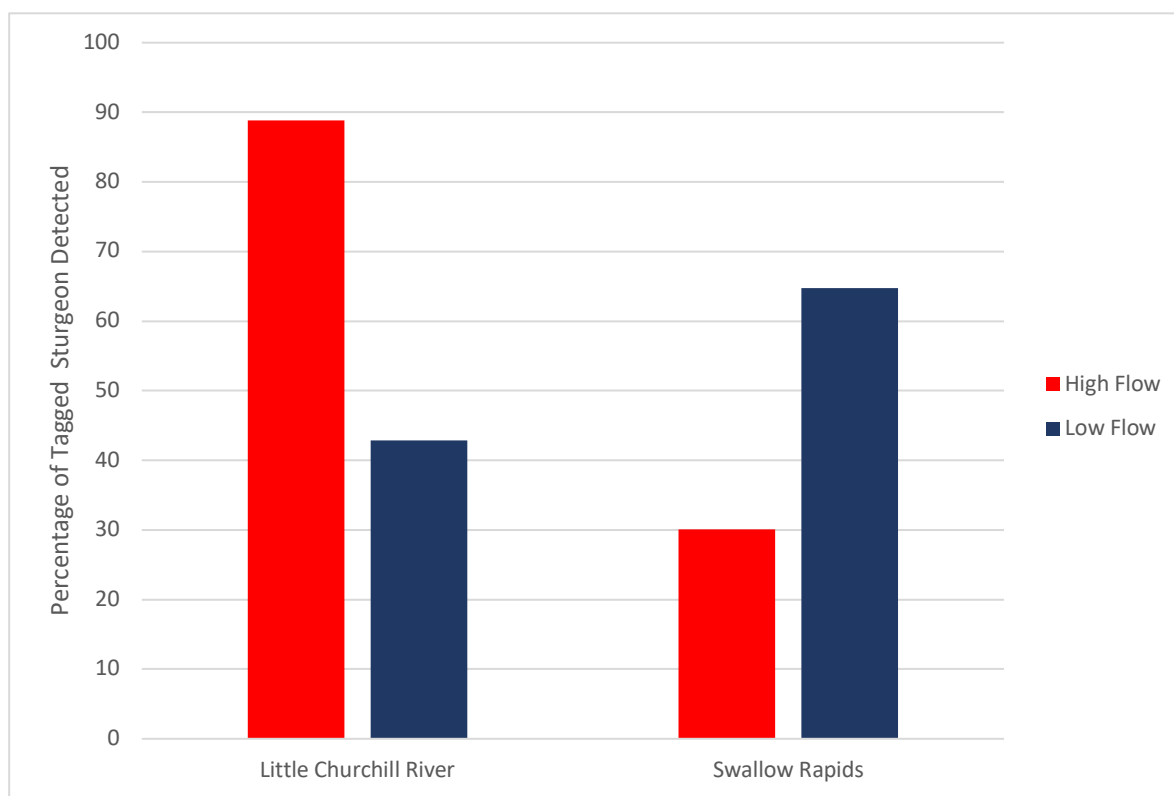


Figure 12. Altered Lake Sturgeon Movement due to flow regime.

The differences between detections during high and low flow years was found to be statistically significant for both locations. The two-sided t-test for the LCR and Swallow rapids gave test statistics of -3.02 and 8.04 respectively, with corresponding p-values of 0.03934 and 0.004016, both lower than the cut-off of 0.05². This suggests that during higher flow years when more areas are accessible to the sturgeon that they prefer to move into the Little Churchill River, while in lower flow years they are stuck to the section of the Churchill River around the Confluence and will either spend more time at Swallow Rapids as a second choice, or inadvertently end up at Swallow Rapids by having few alternative destinations.

3.5 The Confluence and Overwintering Movements

Regardless of the flow condition for a year, or whether or not a sturgeon has made it into the LCR, most sturgeon will head towards the Confluence starting in late July to the beginning of August. By September, nearly all detections are taking place at one of the receivers near the Confluence, as the sturgeon gather there for their overwintering. This movement is shown in Figure 16, with large percentages and number of detections occurring from locations C2 to C4 (greens) towards the end of each summer. It is also depicted in Figure 17 showing the upstream vs downstream movements in both rivers. There is a higher proportion of downstream movement in the LCR, heading towards the Churchill River (red bars), towards the end of July and August. There's no noticeable pattern for movement within the Churchill River (green and yellow bars) as the Confluence is in the middle of the available area to the sturgeon and some swim upstream to it, while others swim downstream.

In 2023, two receivers were placed at C3 – Robert's Camp, and C5 – the Confluence, to see if these locations were where the sturgeon were specifically spending their winters. For most of the winter months fewer than 10 individuals were detected by these two receivers. With sturgeon limiting their movement during the winter months, it is possible that they were somewhere between, or just outside, these two receivers and not detected by either.

² A two-sided t-test is a statistical method used to compare two sets of numbers (in this case, detections during high-flow vs. low-flow years). The results show the difference is unlikely due to chance and is considered statistically significant.

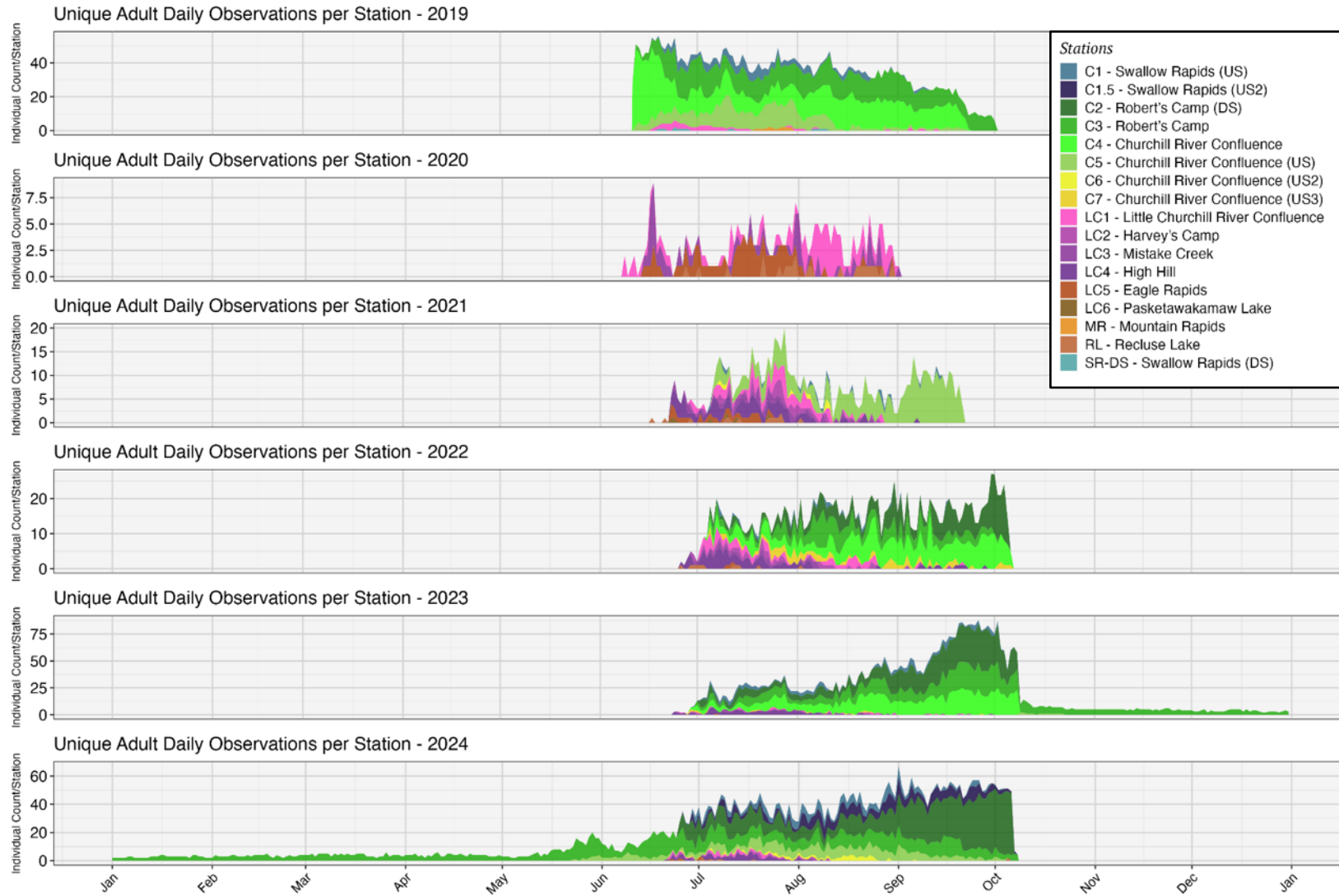


Figure 13. Population movement of Lake Sturgeon from 2019-2024. Each unique individual detected each day adds one to the count for that station. Y-axis adjusted year to year for visual clarity.

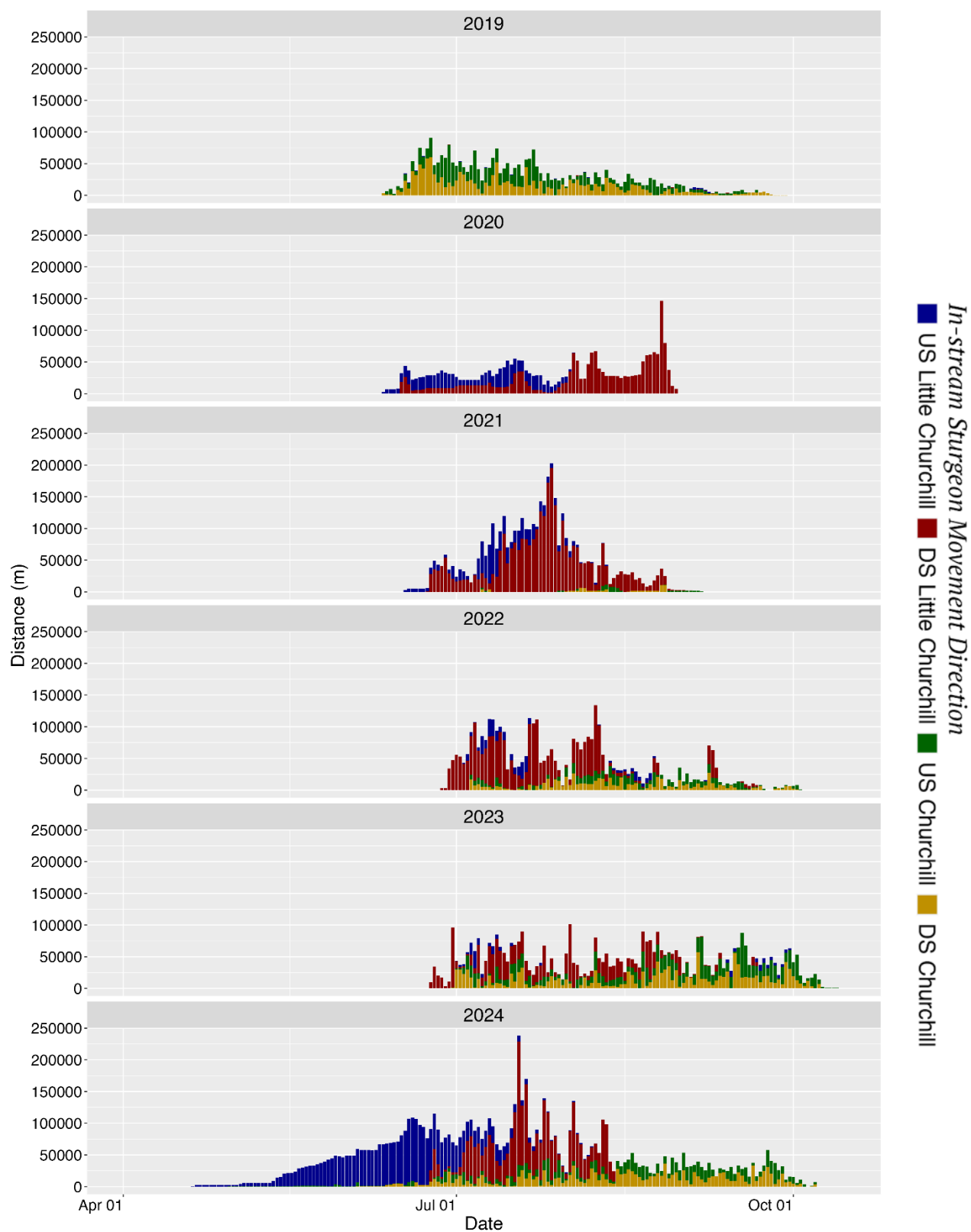


Figure 14. Timing of upstream and downstream movements of Lake Sturgeon in the Churchill and Little Churchill Rivers. US – Upstream, DS – Downstream.

3.6 Seasonal Peaks in Movement

Figures 13 and 14 illustrate where Lake Sturgeon were moving, but the results show that seasonality – especially mid-July peaks – was the dominant driver of movement, whereas discharge had little effect.

After an initial comparison, it was very apparent that there was no correlation to the amount of sturgeon movement in a week with respect to the discharge for that week. While somewhat unexpected, this can be explained by a few reasons. The first is that the receiver locations varied from year to year (Figure 6). With some years having a higher density of receivers in the Churchill River than others, a greater amount of small back-and-forth movements were detected creating the appearance of increased total sturgeon movement compared to years where the receivers were more spread out. This variation is artificial and not related to the actual flow amount. While a more consistent deployment of receivers could have potentially helped determine the relationship of smaller movement patterns to discharge, there was a more important alternate factor.

The time of year seemed to be the main driver of the cumulative distance travelled. Figure 15 below shows the weekly movement sum for each week of the open water season when receivers were set, and Figure 16 shows the average across all years.

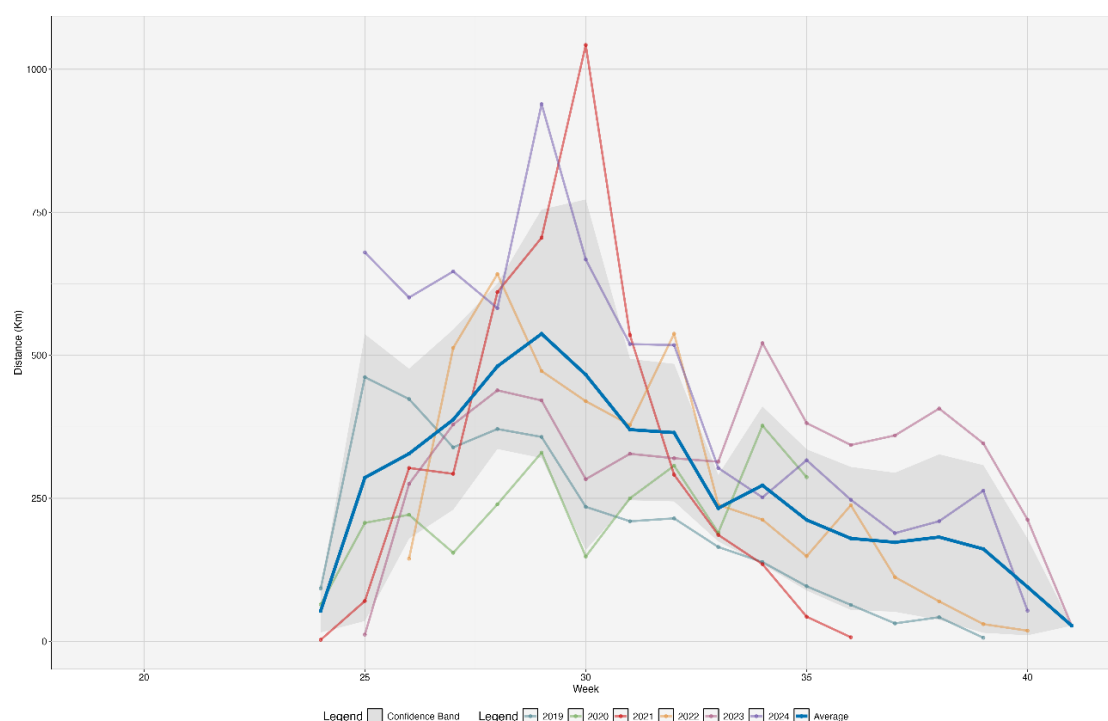


Figure 15. Weekly Lake Sturgeon cumulative travel distance during the open water season. All detected movements for all sturgeon were summed for each week of each year.

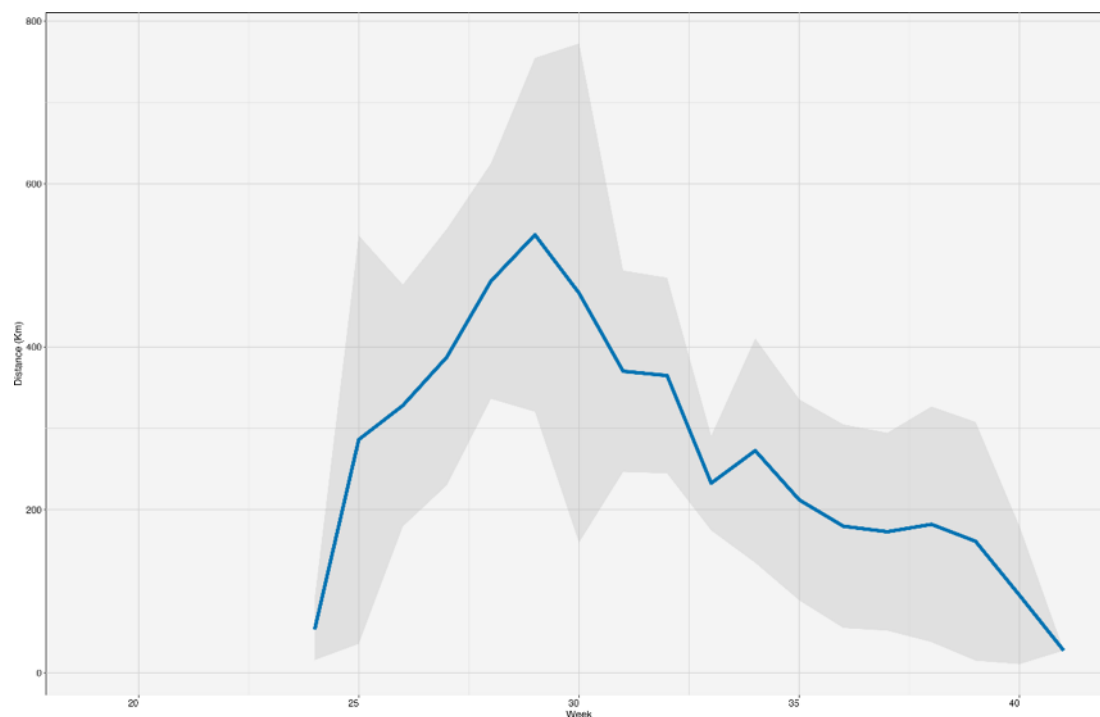


Figure 16. Average weekly cumulative travel distance of Lake Sturgeon. The cumulative movement of all Lake Sturgeon for each week was summed for each year, and the totals for each corresponding week across all years were averaged.

There is a clear peak in the average weekly movement from the 28th to the 30th weeks of the year. These weeks are the 2nd, 3rd, and 4th weeks of July. This is the middle of summer when sturgeon are expected to be active. It is also likely that there is a peak in movement in the spring during migration up the LCR before the receivers are placed and is not captured in the data. Factors such as daylight hours and water temperature are known to increase fish activity and are likely behind the increase in activity. It is interesting to note that while active, it appears that the sturgeon will move similar amounts (so far as the receiver set can distinguish) during high and low flow years, with longer range movements down the LCR during high flow years, and many shorter movements within the Churchill River during low flow years. The peak in movement also appears to loosely correlate with the shift in movement toward the Confluence for the fall.

3.7 Influence of Water Temperature

The final influential factor shaping sturgeon movement is water temperature, with cooler periods in July shifting peaks in activity and declining temperatures in August aligning with reduced movement.

While acute or local changes in water temperature may affect sturgeon's movement, the deployment of receivers was designed to capture long range patterns and does not have the

resolution to observe potential changes from small variations in temperature. However, the effect from large temperature differences was still observable. For instance, the peak movement in 2021 was in the 30th week, which is slightly delayed compared to other years. In Figure 19, showing the temperature in the LCR and Churchill River across the study years, there is a noticeable dip in temperature at the beginning of July which could be responsible for the shift in movement. Also, for 2022, there appeared to be two peaks in movement and a similar pattern in temperature is seen for that year's temperature. Finally, for all years there is a downward trend in temperature that starts near the beginning of August which could be partially behind the decreasing trend in movement during the same time period.

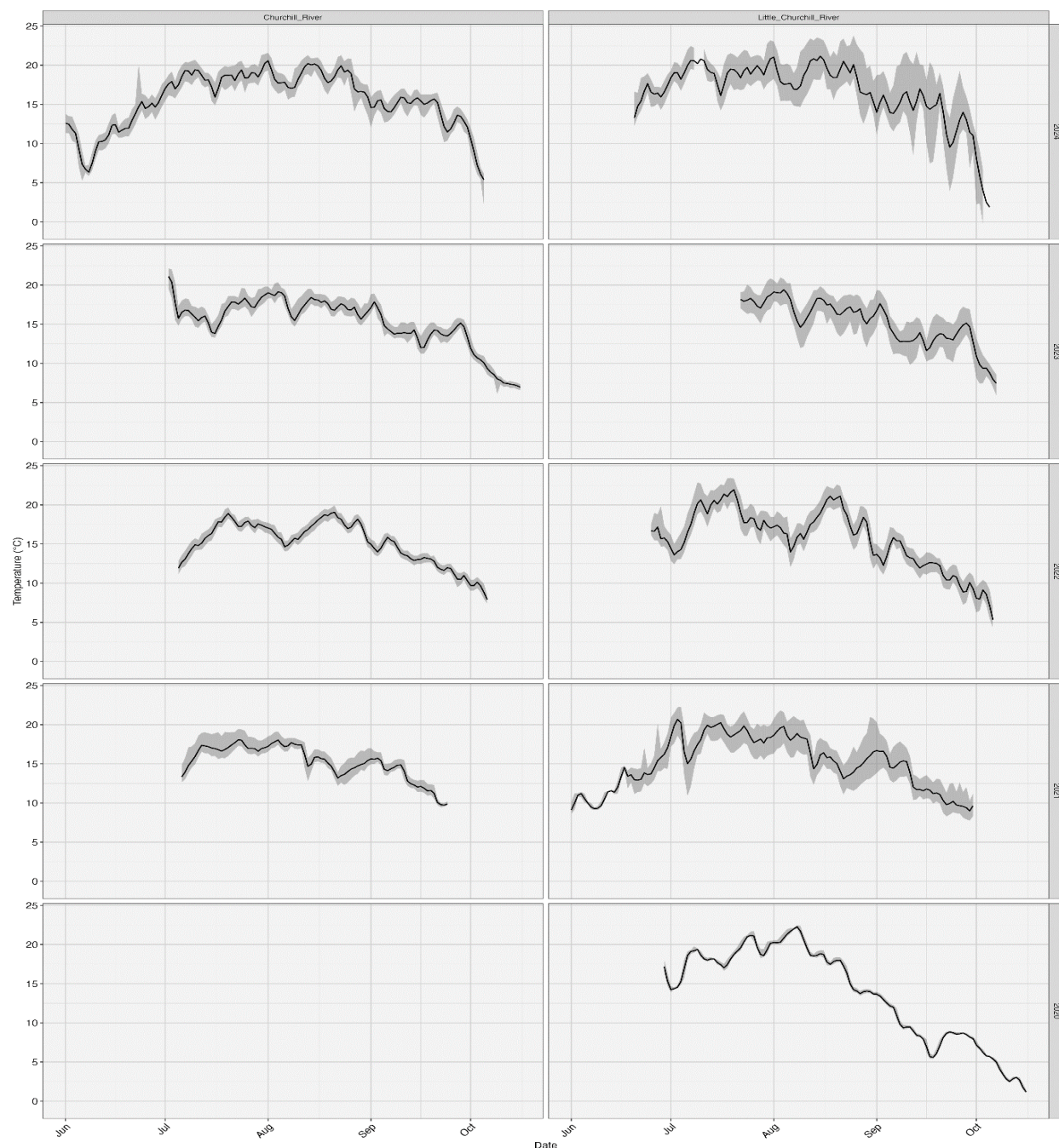


Figure 17. Water temperature in the Churchill (left) and Little Churchill (right) Rivers. Temperatures recorded on Vemco receivers during the study period from 2020 (bottom) – 2024 (top).

3.8 Tag Retention

In 2019, the 47 fish tagged in June were detected within the receiver array during the study period from June 10 to September 30. With the exception of #7, all other fish appeared to have successfully recovered from the transmitter implantation surgery and nearly all

continued to survive and be detected through to the end of 2024. Between the receiver sets in 2021 and 2022, fish #11 was accidentally captured in community nets set by TCN members and was harvested. The transmitter was identified and returned. Fish #24 went missing between the same two years and was not detected again. This fish could have perished, or alternatively, have left the study area. While inside the receiver array, fish #24 was one of the further travelling fish, being detected at Recluse Lake late in the year (end of August) in 2020, and even further at Pasketawakamaw Lake in 2021. This fish returned to the Confluence at the end of 2021, but with the history of distant travels, may have left the study area in the spring of 2021 before receivers were deployed. Either way, fish #24, along with #7 and #11, were considered as lost to the study and percentages for the corresponding years were adjusted accordingly.

4 Discussion

This study found that Lake Sturgeon movements were strongly shaped by flow conditions, seasonality, and water temperature. Most importantly, high flows allowed fish to access and remain in the Little Churchill River (LCR) throughout the summer, while low flows restricted access and forced them to remain in the Churchill River, often at less suitable sites like Swallow Rapids. This major difference in habitat use highlights the importance of adequate flows to sustain sturgeon populations in the system.

Lake Sturgeon are a highly valued and iconic species in North America, but their populations have declined significantly in recent decades due to habitat degradation, overfishing, and other anthropogenic activities (Haxon and Cano, 2016). The Western Hudson Bay population of Lake Sturgeon was subject to changes brought about by anthropogenic activities in the form of hydroelectric development in the region. Such development saw the construction of generating stations and other control structures, like the Missi Falls Control Structure (MFCS), which are barriers to long range movements of sturgeon which they often need to make for fulfillment of their various life stages. COSEWIC reports that the degradation and fragmentation of Lake Sturgeon habitat that has ultimately led to their listing of “endangered” across Manitoba is largely a result of dams (COSEWIC, 2017).

This study focused on the Churchill River near the confluence with the LCR, and the LCR itself, where the last sizeable population of Lake Sturgeon in the management area resides (COSEWIC, 2017) following widespread habitat change after the completion of the MFCS.

Given the massive dewatering effect the MFCS has had on the Lower Churchill River, this study into the effects of flow, water temperature, and seasonality on Lake Sturgeon movement is a necessary step to understand what, if any, requirements in environmental conditions exist in order for Lake Sturgeon to access the habitat they require. This study represents the first of its kind in the region.

The first obvious trend in the data was the reliance on the area of the Confluence, which was heavily used throughout the entire study. The Confluence is an area where the Churchill River widens as the LCR feeds into it. It is an area of reduced flow and is a likely spot for sediment deposition, favourable for the benthic organisms that Lake Sturgeon feed on. The reduced flow may also make the region desirable for overwintering as overwintering locations are often of low flows.

Finding heavy use of this area is consistent with previous studies of adult sturgeon movement. Knights et al. (2002) noted that core areas of sturgeon use were hydrologically similar, commonly involving transition areas from high to low flow. Similarly, Struthers et al. (2017) studied Lake Sturgeon on the Winnipeg River, another river system impacted by generating stations, and found foraging and overwintering activities near transitions between riverine (flowing river) and lacustrine (lake or reservoir) habitat. Further away, Lake Sturgeon in the St. Mary's River in Michigan were also found to extensively use transition areas between high and low flow waters (Gerig, 2011).

A major difference captured during this study was the altered movement behaviour during low flow versus high flow conditions. It appeared that the cumulative distance travelled did not significantly differ between high and low flow years, but where the Lake Sturgeon travelled, and consequently, the range of their movements did. There was a higher percentage of Lake Sturgeon travelling from the Churchill River into and up the LCR during high flow years over low flow years. During those low flow years, there was a higher percentage travelling downstream in the Churchill River to reach Swallow Rapids. This suggests that during periods of higher flow (and higher water levels) when there are fewer barriers to movement, the Lake Sturgeon are preferentially occupying the LCR. They would continue to occupy the LCR past the spring spawning window indicating that the habitat found within the LCR also supports summer feeding activities. Since the LCR is isolated from changes stemming from the Churchill River Diversion (CRD), the more natural flow regimes may also contribute to the preferential occupation of the LCR.

During low flow years, the increased presence at Swallow Rapids either indicates a second choice for preferred habitat, or alternatively, the Lake Sturgeon are inadvertently spending

more time there as they move within a limited range of the Churchill River. These results highlight the importance of maintaining adequate flows within the Churchill River to ensure continued access to the LCR since Lake Sturgeon will normally travel great distances to reach their required habitat, and there is a clear preference for longer travels up the LCR when available. Prior to the CRD, the discharge minimums were much higher than they are post CRD. Requiring higher minimum flows in the Lower Churchill River similar to pre-CRD levels would likely increase the access and available habitat to Lake Sturgeon and help the remaining population sustain itself and spread further within the watershed.

The LCR is a long stretch of river and connected lakes and represents various habitat types along its expanse. It is noteworthy that Lake Sturgeon, while in the LCR, would spend a disproportionate amount of time at the receiver near High Hill. This is an indication that there is important habitat there for potential spawning or feeding activities. Fish #13 stayed in the vicinity of High Hill for nearly the entire length of observation within this study which is evidence of potentially important habitat in that area. The same fish was detected in or near this location late into each year, after the point when most other individuals have returned to the Confluence which opens the possibility of overwintering habitat in the area. Although, it is still possible the fish may have returned to the Confluence after the receivers were pulled out each year.

The availability of overwintering habitat is a must for continued sturgeon survival. Another population of Lake Sturgeon in the South Saskatchewan River was studied for its overwintering behavior and it was discovered that about a third of the adult population spent the entire winter within a 2 km stretch while 1100 km of habitat was available to them (Thayer, 2017), providing strong evidence for the importance of overwintering habitat.

A similar trend was observed in our study where overwintering was done near the confluence of the Churchill River and LCR. The Confluence is one of the last remaining deep sections of river accessible to the sturgeon and the gathering of Lake Sturgeon in the Confluence, specifically near Robert's camp or just downstream (locations C2 and C3), late in the fall each year indicates that their overwintering habitat is nearby. The exact location is still unknown as most receivers were removed before freeze-up. Receivers were left in the Churchill River at locations C3 and C5 to determine the extent of movement during the winter months, however, minimal detections were made on either indicating two things. One, that the exact location that the Lake Sturgeon use for overwintering was not at these locations, and two, assuming the true location was in fact nearby, there is minimal movement of Lake Sturgeon throughout the winter months.

In areas outside the Confluence, the extremely low flows of the Churchill River in the winter have limited the availability of overwintering habitat. This is likely a key contributing factor to the loss of Lake Sturgeon in those areas.

As flows in the Churchill River have been reduced, the river itself has been minimized. Consequently, the available habitat for Lake Sturgeon to colonize has also been reduced. For the population to continue to survive and recover, adequate food needs to be available. During low flow years when Lake Sturgeon are mostly limited to the small stretch of the Churchill River between Mountain and Swallow Rapids, the available food may not be enough to support a healthy, growing population. Our population assessment (AAE, 2025) investigates the structure of the population and makes several recommendations to this end.

While the reduction of habitat in the Churchill River system is concerning, there may be opportunities for improved utilization of the Little Churchill River if proper flow can be maintained in the Churchill River system. In 2021, three fish (#3, #23, and #24) were detected moving from Eagle Rapids upstream into Pasketewakamaw Lake, the only year a receiver was placed in this lake. The presence of Lake Sturgeon in Pasketewakamaw Lake is a highly significant and unexpected finding, as TCN members previously believed that Lake Sturgeon had been extirpated from this habitat, which had long been used for subsistence fishing. It was previously thought that barriers to fish passage existed at Eagle Rapids and Innis-Taylor Rapids (out of Recluse); however, this study provides evidence that the upper reaches of the LCR can be accessed and utilized under stable flow conditions. If proper flow can be maintained in the Churchill River system, it is possible that Lake Sturgeon may recolonize this historically important habitat.

4.1 Study Limitations

This study used a wide spread of receivers deployed on average from late June to the end of September/beginning of October. It appeared that most Lake Sturgeon movement up the LCR occurred sometime before that as the most fish found in the LCR were detected as soon as, or shortly after, receivers were deployed. Without recording the movement on the receivers, it is impossible to determine any impactful factors that may alter the timing or nature of the spring movement. Additionally, by pulling receivers out by the fall, the Lake Sturgeon's location and movements during the winter remain unconfirmed beyond the fact they are somewhere near the Confluence and exhibit low movement.

A wide range of locations were used throughout the study, and the locations shifted from year to year. This allowed for the determination of preferred areas, such as High Hill, that

would have been missed with a more localized, dense set. However, the spacing of the receivers lacked data on finer scale movements which could potentially correlate with factors such as daily fluctuations in temperature or flow. Comparisons between years using movement data becomes less meaningful as the distance between the receivers and their locations will impact to what extent movements are detected. For this reason, a large part of the analysis within this study focused on general trends or reduced detections to a binary level of analysis (were they detected, or not detected, in a given location in a given year).

There were also a few gaps in the data created by extraneous circumstances. Throughout the Lower Churchill River below Missi Falls, intense currents, large standing waves, floating debris and submerged hazards in 2020 created an environment unfit for safe navigation. As a result, all receivers in the Churchill River could not be retrieved that year. The attempted retrieval of these receivers in 2021 was unsuccessful. It is probable that increased sedimentation rates caused by the dramatic increase in flows in 2020 buried the receivers beneath the sediment. Also, in 2021, receiver LC2, set within the Little Churchill River by Harvey's Camp, was found removed from the river by wildlife. The receiver was pulled up on the riverbank and out of the water. It is unlikely that the receiver was removed from the river until sometime during the fall since detections were recorded into late August. In all likelihood, detections were likely missed once the receiver was removed from the river.

4.2 Future Directions

To continue addressing the knowledge gaps identified and further corroborate conclusions drawn from studies of sturgeon movement conducted from 2019-2024, AAE proposes to continue monitoring the movement, dispersal, and distribution of Lake Sturgeon in the river system across several years. As the batteries will have run out on the transmitters used in this study, a new set of Lake Sturgeon would need to be tagged and would be an opportunity to increase the sample size (increase tag saturation). A greater number of receivers could be deployed to increase coverage, maximizing the temporal and spatial resolution in telemetry results and increasing the strength of conclusions based on these data. Receivers should be deployed earlier to capture initial movement and timing into the LCR in the spring as well as additional receivers left over winter to better locate the sturgeon's overwintering grounds.

Building on the strong evidence of critical habitats identified in this study, additional investigations into habitat characteristics – such as the development of habitat suitability indices – would further refine understanding of the environmental conditions that support a thriving sturgeon population in this system. This could include more detailed assessment of spawning habitat at traditional (pre-CRD) locations that have been identified by TCN

elders and local resource users, including the Little Beaver River and tributaries to the Little Churchill River, alongside potential spawning locations identified in this study, such as High Hill. Larval sturgeon occurrence at these locations could provide additional verification that the areas are actively used for spawning. Further studies could also assess the habitat used by juveniles. Ongoing investigations of juvenile sturgeon movement throughout the river system will help characterize areas of high usage and the habitat features they rely on.

Other studies focused on changes brought on by the CRD can help determine the extent of its impact. These studies can include examining the impacts of increased bank erosion to high quality spawning habitat along the LCR. The analysis of benthic organisms that would be impacted first by changes in water quality could be an appropriate means to accomplish this task. These studies could also include expanding the areas of investigation to include South Indian Lake and others upstream of Missi Falls.

Finally, actions that could be taken to help the recovery of Lake Sturgeon in the Churchill River could be studied. This could include a feasibility study to possible stocking upstream of Mountain Rapids. It appears that Mountain Rapids was a barrier to upstream movement of the sturgeon and stocking beyond this point could help recolonize further reaches of the river system. The establishment of new minimum flow regulations within the Lower Churchill River could help provide expanded winter habitat crucial for Lake Sturgeon survival and also help ensure access to preferred areas such as the Little Churchill River.

5 Acknowledgements

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Appendix A1 – Supplemental Information

Table A 1. Detailed information on each tagged individual Lake Sturgeon.

Tag ID	Sturgeon ID	Fork Length (mm)	Total Length (mm)	Weight (Kg)	Maturity	Sex	Age ATOC	Hatch Year	Age (2024)
7680	A1	811	928	4.8	Immature	Unk.	15	2004	20
7679	A2	1009	1014	7.4	Immature	Unk.	24	1995	29
7678	A3	875	935	5.3	Immature	Unk.	19	2000	24
7675	A4	1321	1447	24.2	Mature	Female	41	1978	46
7676	A5	1150	1259	11.2	Immature	Unk.	NA	NA	NA
7677	A6	965	1175	7.2	Immature	Unk.	27	1992	32
7659	A7	1456	1562	27.9	Mature	Female	24	1995	29
7658	A8	1041	1146	9.4	Immature	Unk.	37	1982	42
7657	A9	1072	1199	9.9	Immature	Unk.	29	1990	34
7662	A10	1140	1200	19.4	Mature	Unk.	22	1997	27
7661	A11	1242	1375	17.5	Mature	Female	38	1981	43
7660	A12	1089	1179	9.2	Unknown	Unk.	24	1995	29
7665	A13	1050	1074	9.9	Immature	Unk.	18	2001	23
7667	A14	1100	1210	9.8	Immature	Unk.	37	1982	42
7666	A15	1300	1440	17.7	Mature	Female	45	1974	50
7668	A16	1081	1195	9	Mature	Male	34	1985	39
7664	A17	868	961	7.1	Immature	Unk.	18	2001	23
7663	A18	1011	1140	7.6	Immature	Unk.	26	1993	31
7671	A19	1211	1310	12.7	Immature	Unk.	34	1985	39
7670	A20	951	1080	6.4	Immature	Unk.	25	1994	30
7669	A21	941	1052	6.39	Immature	Unk.	21	1998	26
7674	A22	1100	1340	7.6	Mature	Male	27	1992	32
7673	A23	1071	1192	9	Immature	Unk.	32	1987	37
7672	A24	1356	1483	17.3	Mature	Male	31	1988	36
7641	A25	1177	1298	12.9	Mature	Male	36	1983	41
7640	A26	1330	1422	18.2	Mature	Unk.	24	1995	29
7639	A27	1034	1160	8.3	Immature	Unk.	28	1991	33
7642	A28	1050	1152	9.7	Immature	Unk.	30	1989	35
7643	A29	1415	1473	27.6	Mature	Female	NA	NA	NA
7644	A30	900	1025	5.8	Immature	Unk.	19	2000	24
7645	A31	1020	1100	8.7	Mature (R+R)	Male	29	1990	34
7646	A32	1165	1293	13.7	Immature	Female	16	2003	21
7647	A33	955	1067	7.3	Immature	Unk.	17	2002	22
7648	A34	841	940	5.6	Immature	Unk.	NA	NA	NA
7649	A35	1142	1273	10.5	Immature	Unk.	25	1994	30
7650	A36	1219	1335	15.1	Mature	Unk.	38	1981	43
7651	A37	1410	1550	22.1	Mature	Female	48	1971	53

7652	A38	1423	1552	20.2	Mature (R+R)	Male	41	1978	46
7653	A39	1191	1311	13.3	Mature (R+R)	Male	NA	NA	NA
7654	A40	1084	1190	8.9	Mature	Male	NA	NA	NA
7655	A41	930	1078	7.3	Immature	Unk.	1	2017	7
7656	A42	1190	1317	10.8	Mature	Male	NA	NA	NA
7636	A43	1080	1185	9	Immature	Unk.	35	1984	40
7637	A44	1148	1275	11.1	Immature	Unk.	NA	NA	NA
7638	A45	925	1043	6.5	Mature	Male	NA	NA	NA
7633	A46	940	1046	6.7	Immature	Unk.	20	1999	25
7634	A47	1091	1232	10.3	Mature	Male	NA	NA	NA

Table A 2. 2019 Vemco Receiver deployment information.

Year	Receiver ID	Code and Location	Latitude	Longitude	Set Date	Pull Date
2019	482582	C1 - Swallow Rapids (US)	57.57045	-95.43958	June 10 2019	Sept 30 2019
2019	482594	LC1 - Little Churchill River Confluence	57.50508	-95.35972	June 11 2019	Sept 30 2019
2019	482589	C5 - Churchill River Confluence (US)	57.51413	-95.39592	June 10 2019	Sept 30 2019
2019	482595	C4 - Churchill River Confluence	57.5112	-95.36117	June 9 2019	Sept 30 2019
2019	482591	BL - Billard Lake	57.14908	-96.1268	July 16 2019	Sept 30 2019
2019	482592	BL-DS - Billard Lake (DS)	57.15853	-95.97945	July 19 2019	Sept 30 2019
2019	481267	CW1 - Churchill Weir (US)	58.33438	-94.31893	June 26 2019	Sept 30 2019
2019	481261	CW - Churchill Weir	58.6693	-94.18347	June 25 2019	Sept 30 2019
2019	482593	MR - Mountain Rapids	57.36363	-95.69402	July 19 2019	Sept 30 2019
2019	482584	C3 - Robert's Camp	57.53383	-95.32863	June 9 2019	Sept 30 2019
2019	482587	SR-DS - Swallow Rapids (DS)	57.61233	-95.43932	June 14 2019	Sept 30 2019
2019	482597	LC1 - Little Churchill River Confluence	57.50098	-95.36234	June 14 2019	Sept 30 2019

Table A 3. 2020 Vemco Receiver deployment information.

Year	Receiver ID	Code and Location	Latitude	Longitude	Set Date	Pull Date
2020	482583	C4 - Churchill River Confluence	57.510268	-95.376075	June 7 2020	NA
2020	482593	C7 - Churchill River Confluence (US3)	57.481402	-95.493456	June 9 2020	NA
2020	482595	C1 - Swallow Rapids (US)	57.569985	-95.440809	June 9 2020	NA
2020	482596	LC1 - Little Churchill River Confluence	57.501338	-95.362558	June 7 2020	Sept 7 2020
2020	482578	LC3 - Mistake Creek	57.178968	-95.345266	June 7 2020	Sept 7 2020
2020	481261	LC5 - Eagle Rapids	56.959549	-95.665329	June 7 2020	Sept 7 2020
2020	482582	RL - Recluse Lake	56.938579	-95.695614	June 28 2020	Oct 1 2021
2020	482581	LBR - Little Beaver River	57.741159	-95.481054	June 7 2020	NA
2020	482579	BL - Billard Lake	57.149075	-96.126805	June 7 2020	NA

Table A 4. 2021 Vemco Receiver deployment information.

Year	Receiver ID	Code and Location	Latitude	Longitude	Set Date	Pull Date
2021	482597	C5 - Churchill River Confluence (US)	57.50858	-95.38572	July 5 2021	Oct 2 2021
2021	482578	LC5 - Eagle Rapids	56.95937	-95.66611	June 16 2021	Oct 2 2021
2021	482596	LC7 - Hale Lake	56.63329	-96.14258	June 16 2021	Oct 2 2021
2021	482592	LC2 - Harvery's Camp	57.37408	-95.37248	June 16 2021	Oct 2 2021
2021	482594	LC4 - High Hill	57.12481	-95.49848	June 16 2021	Oct 2 2021
2021	482584	LC1 - Little Churchill River Confluence	57.50126	-95.36254	June 16 2021	Oct 2 2021
2021	482589	LC3 - Mistake Creek	57.17339	-95.34428	June 16 2021	Oct 2 2021
2021	482585	LC6 - Pasketawakamaw Lake	56.78422	-95.79199	June 16 2021	Oct 2 2021
2021	482587	C1 - Swallow Rapids (US)	57.57265	-95.43554	July 5 2021	Oct 2 2021
2021	481267	C6 - Churchill River Confluence (US2)	57.51494	-95.42728	July 5 2021	Oct 2 2021

Table A 5. 2022 Vemco Receiver deployment information.

Year	Receiver ID	Code and Location	Latitude	Longitude	Set Date	Pull Date
2022	482596	C4 - Churchill River Confluence	57.511034	-95.369735	July 4 2022	Oct 6 2022
2022	482578	LC5 - Eagle Rapids	56.959881	-95.664874	June 24 2022	Oct 6 2022
2022	482584	C2 - Robert's Camp (DS)	57.563266	-95.337295	July 4 2022	Oct 6 2022
2022	482592	LC2 - Harvery's Camp	57.37352	-95.371881	June 24 2022	Oct 6 2022
2022	482594	LC4 - High Hill	57.124842	-95.498098	June 24 2022	Oct 6 2022
2022	482585	LC1 - Little Churchill River Confluence	57.501058	-95.361519	June 24 2022	Oct 6 2022
2022	482589	LC3 - Mistake Creek	57.174088	-95.344671	June 24 2022	Oct 6 2022
2022	482597	C3 - Robert's Camp	57.538549	-95.330222	July 4 2022	Oct 6 2022
2022	482582	RL - Recluse Lake	56.939743	-95.699692	June 24 2022	Oct 6 2022
2022	482579	C1 - Swallow Rapids (US)	57.56941	-95.449646	July 4 2022	Oct 6 2022
2022	481267	C7 - Churchill River Confluence (US3)	57.448682	-95.568334	July 4 2022	Oct 6 2022

Table A 6. 2023 Vemco Receiver deployment information.

Year	Receiver ID	Code and Location	Latitude	Longitude	Set Date	Pull Date
2023	482584	C4 - Churchill River Confluence	57.511111	-95.360833	July 1 2023	Oct 9 2023
2023	482578	LC5 - Eagle Rapids	56.959722	-95.664722	June 21 2023	Oct 9 2023
2023	482582	C3 - Robert's Camp	57.528889	-95.331111	July 1 2023	Oct 9 2023
2023	482592	LC2 - Harvery's Camp	57.373611	-95.3725	June 21 2023	Oct 9 2023
2023	482594	LC4 - High Hill	57.125	-95.498611	June 21 2023	Oct 9 2023
2023	482585	LC1 - Little Churchill River Confluence	57.490833	-95.360833	June 21 2023	Oct 9 2023
2023	482589	LC3 - Mistake Creek	57.174167	-95.344444	June 21 2023	Oct 9 2023
2023	482596	C2 - Robert's Camp (DS)	57.5675	-95.344444	July 1 2023	Oct 9 2023
2023	482579	C1 - Swallow Rapids (US)	57.57	-95.447778	July 1 2023	Oct 9 2023
2023	482597	C7 - Churchill River Confluence (US3)	57.468889	-95.506111	July 1 2023	Oct 9 2023

Table A 7. 2024 Vemco Receiver deployment information.

Year	Receiver ID	Code and Location	Latitude	Longitude	Set Date	Pull Date
2023	482579	C3 - Robert's Camp	57.53227	-95.33006	Oct 9 2023	Nov 11 2024
2023	481267	C5 - Churchill River Confluence (US)	57.50984	-95.39195	Oct 8 2023	Nov 11 2024
2024	482597	RL - Recluse Lake	56.93828	-95.69778	June 19 2024	Nov 11 2024
2024	482596	LC5 - Eagle Rapids	56.95946	-95.66604	June 19 2024	Oct 7 2024
2024	482585	LC4 - High Hill	57.12481	-95.49898	June 20 2024	Oct 7 2024
2024	482592	LC2 - Harvery's Camp	57.37362	-95.37144	June 21 2024	Oct 7 2024
2024	482594	LC1 - Little Churchill River Confluence	57.50098	-95.36234	June 21 2024	Nov 11 2024
2024	482578	C1 - Swallow Rapids (US)	57.57258	-95.43616	June 26 2024	Nov 11 2024
2024	482589	C6 - Churchill River Confluence (US2)	57.51516	-95.42786	June 25 2024	Nov 11 2024
2024	482582	C2 - Robert's Camp (DS)	57.56757	-95.34697	June 25 2024	Nov 11 2024
2024	481261	C1.5 - Swallow Rapids (US2)	57.56151	-95.39322	June 25 2024	Nov 11 2024

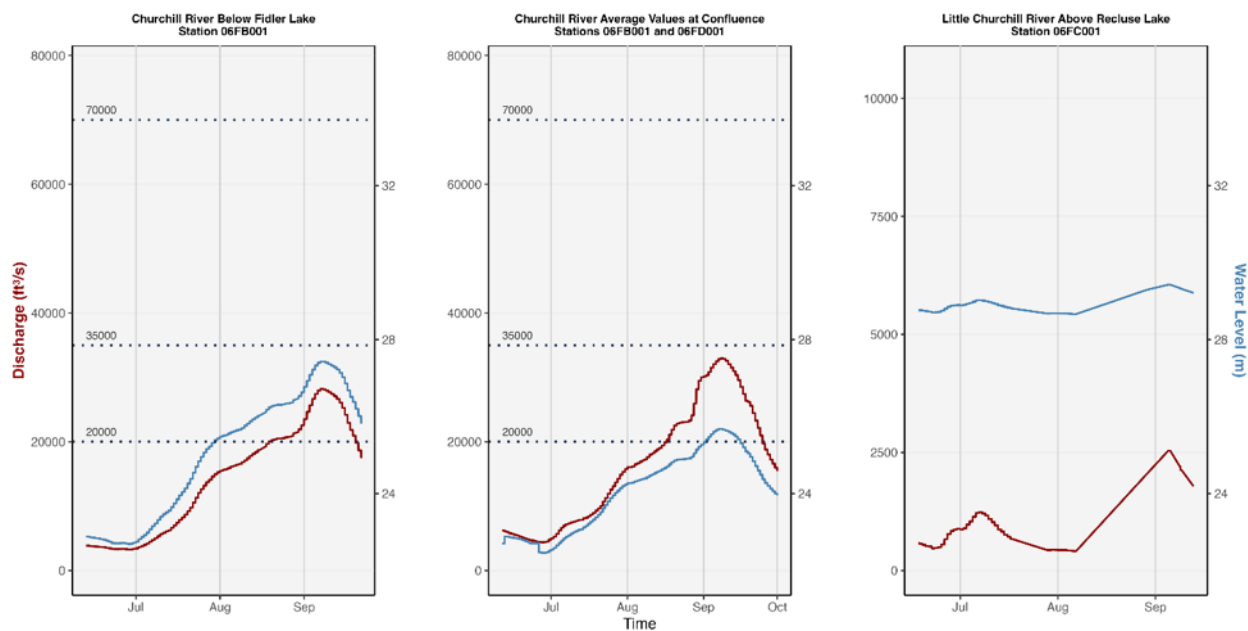


Figure A 1. 2019 Water level discharge in Churchill and Little Churchill Rivers.

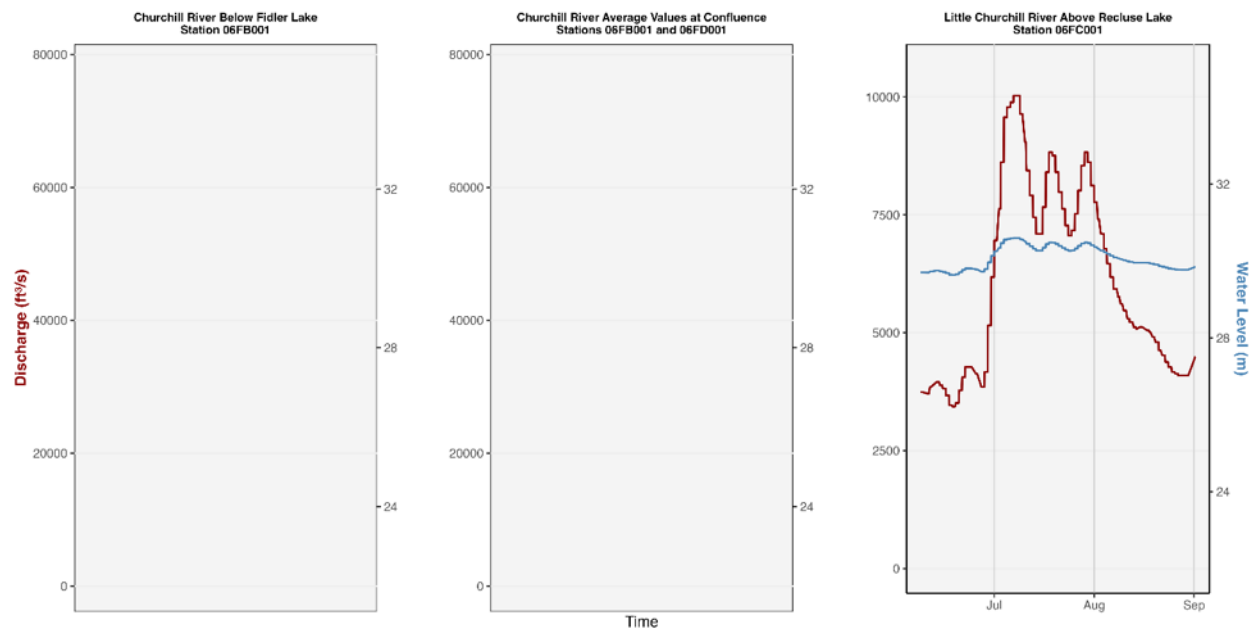


Figure A 2. 2020 Water level discharge in Churchill and Little Churchill Rivers.

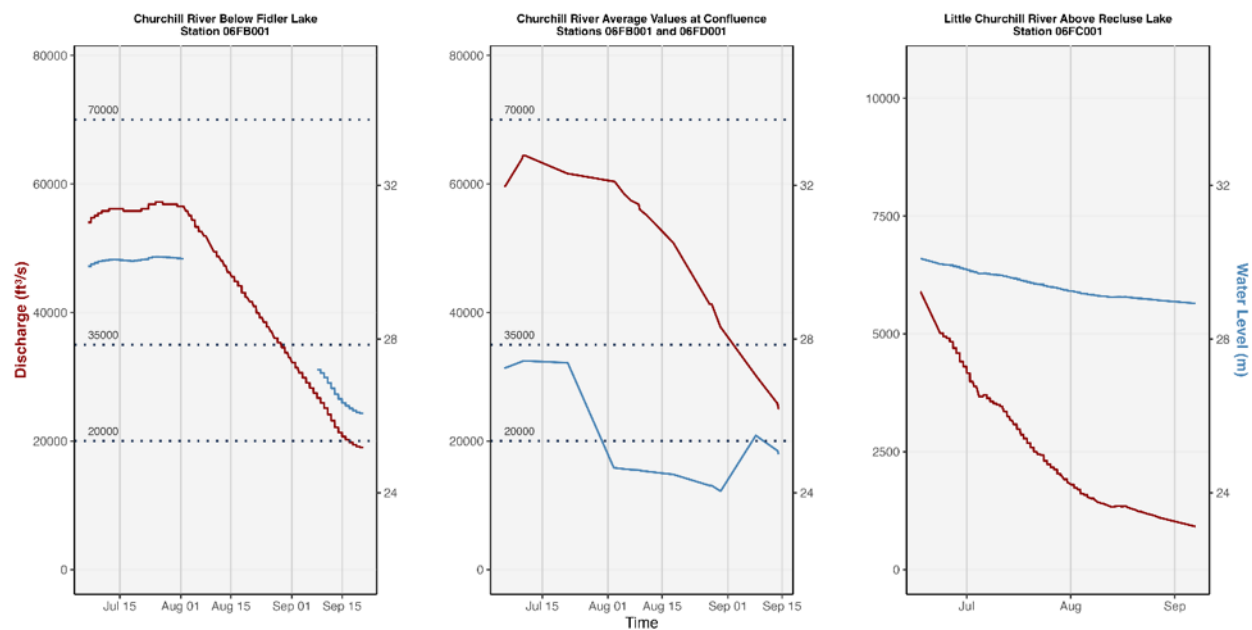


Figure A 3. 2021 Water level discharge in Churchill and Little Churchill Rivers.

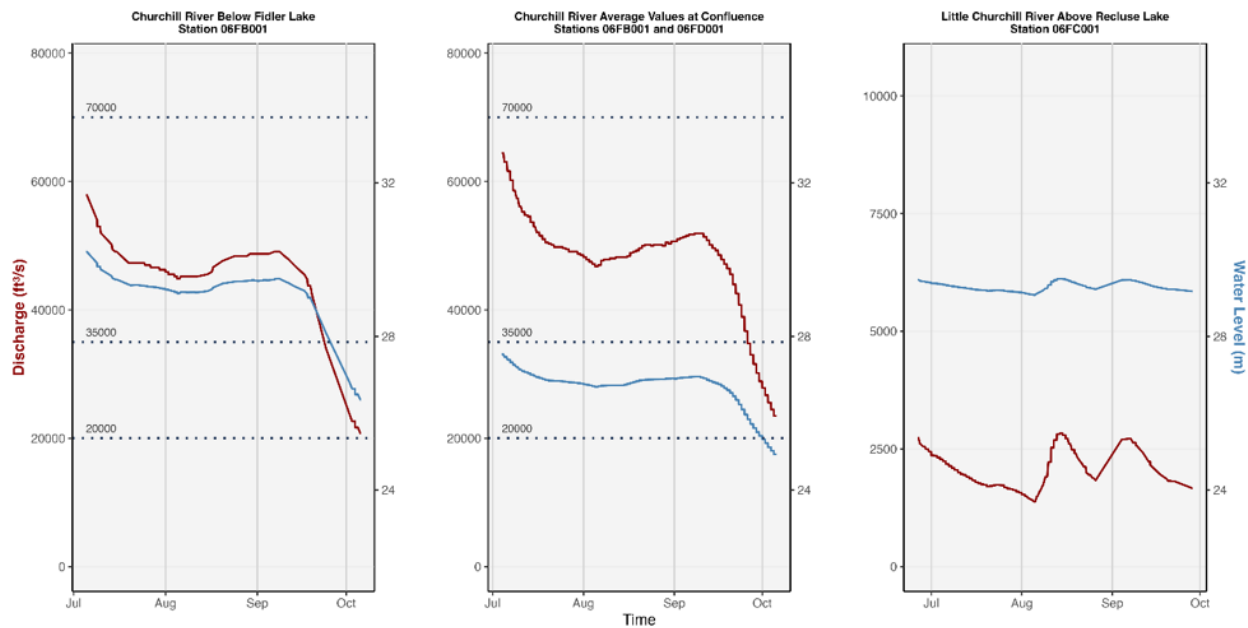


Figure A 4. 2022 Water level discharge in Churchill and Little Churchill Rivers.

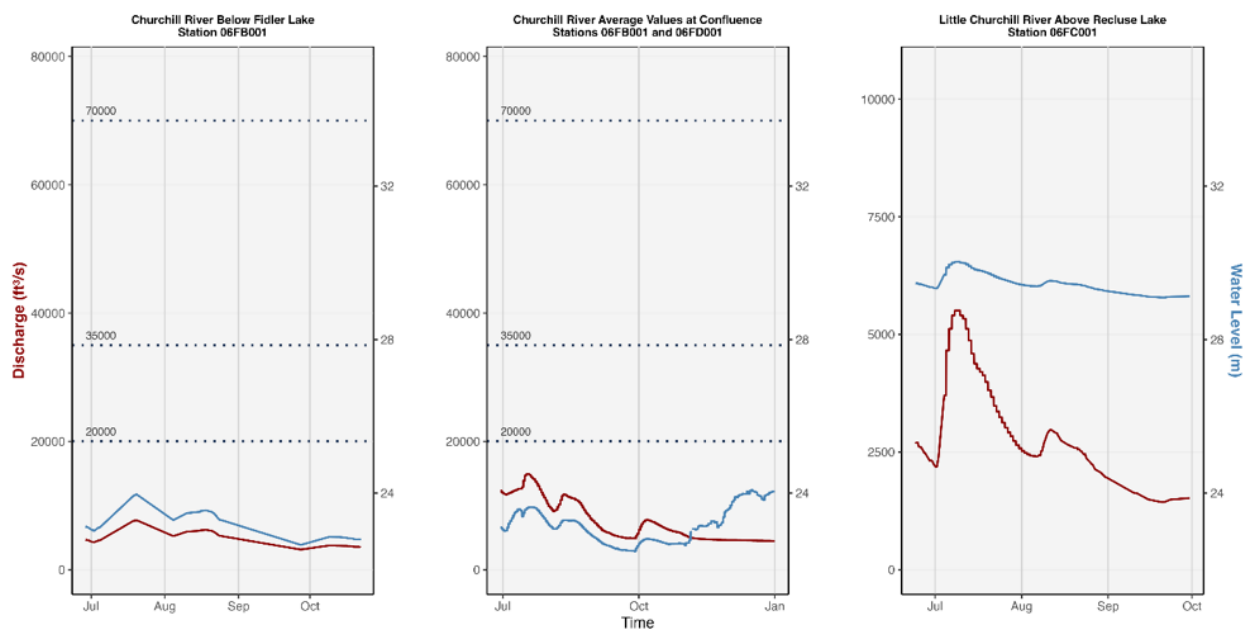


Figure A 5. 2023 Water level discharge in Churchill and Little Churchill Rivers.

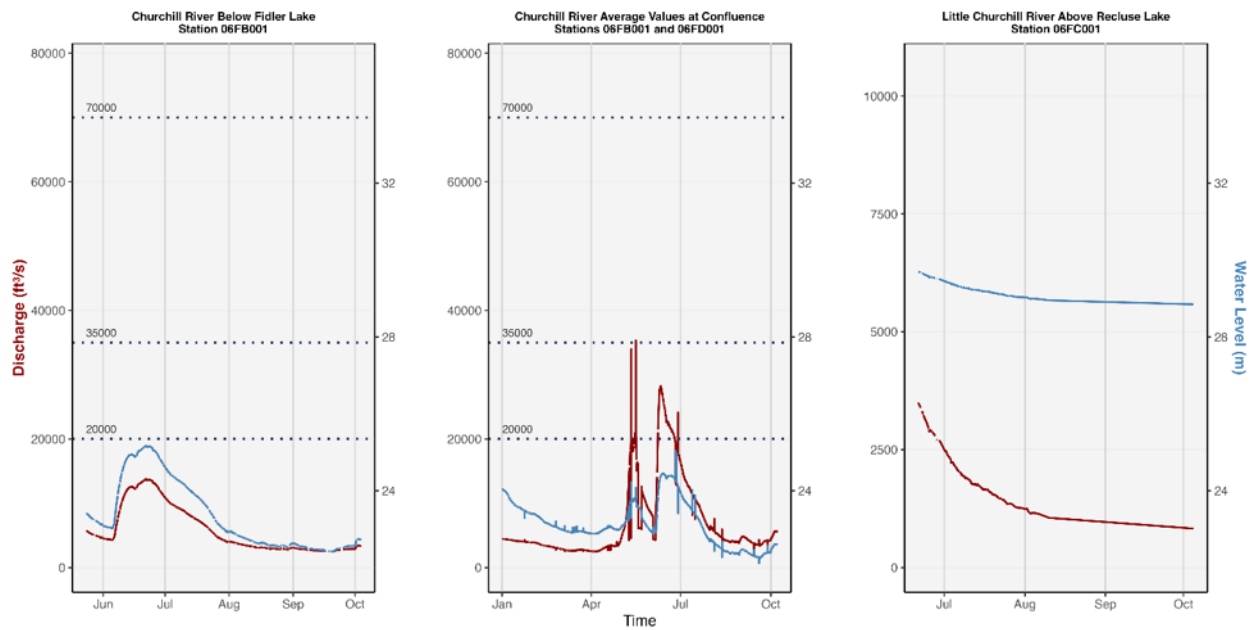


Figure A 6. 2024 Water level discharge in Churchill and Little Churchill Rivers.

Appendix A2 – Individual Sturgeon Movement

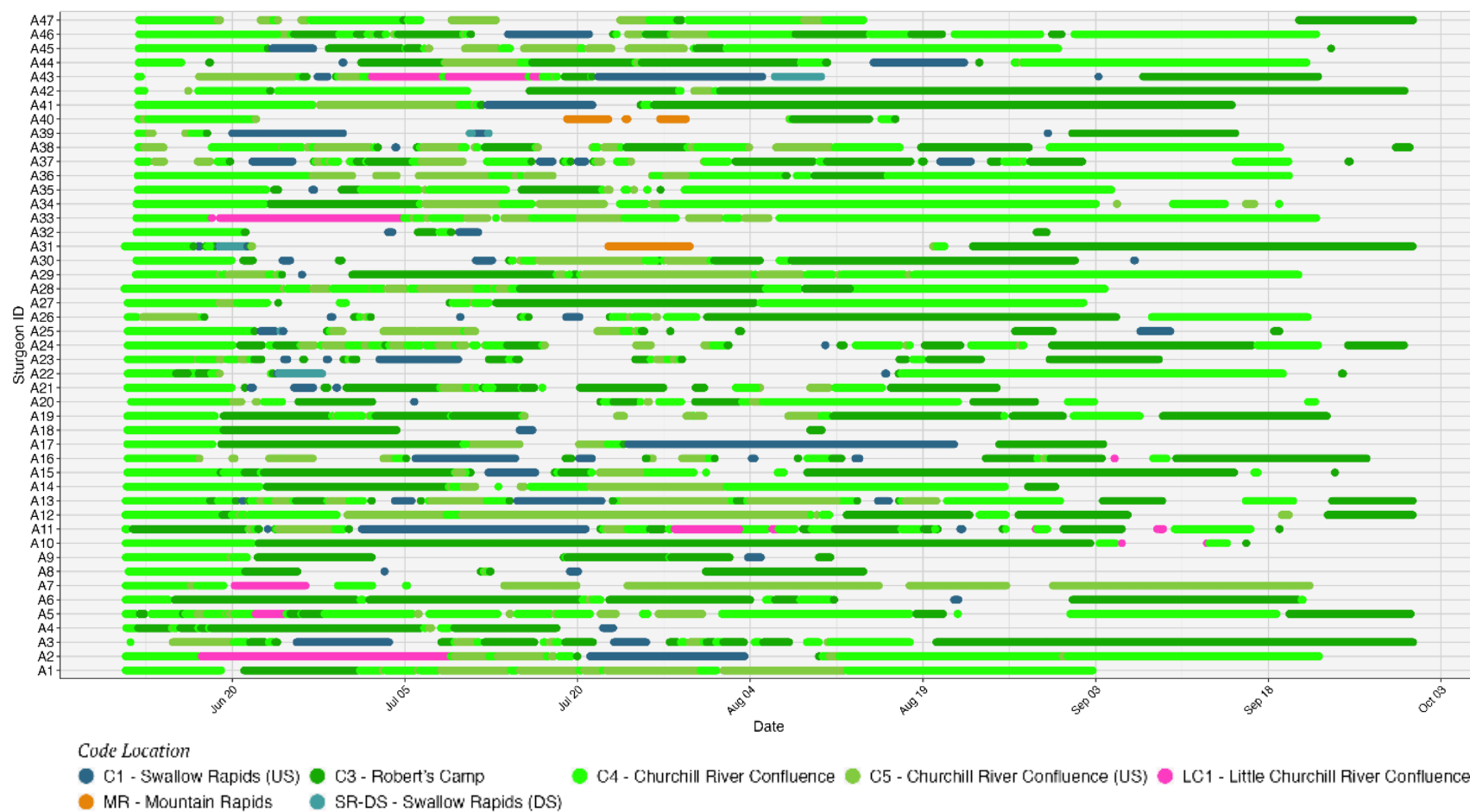


Figure A 7. Individual Lake Sturgeon movements tracked in 2019.

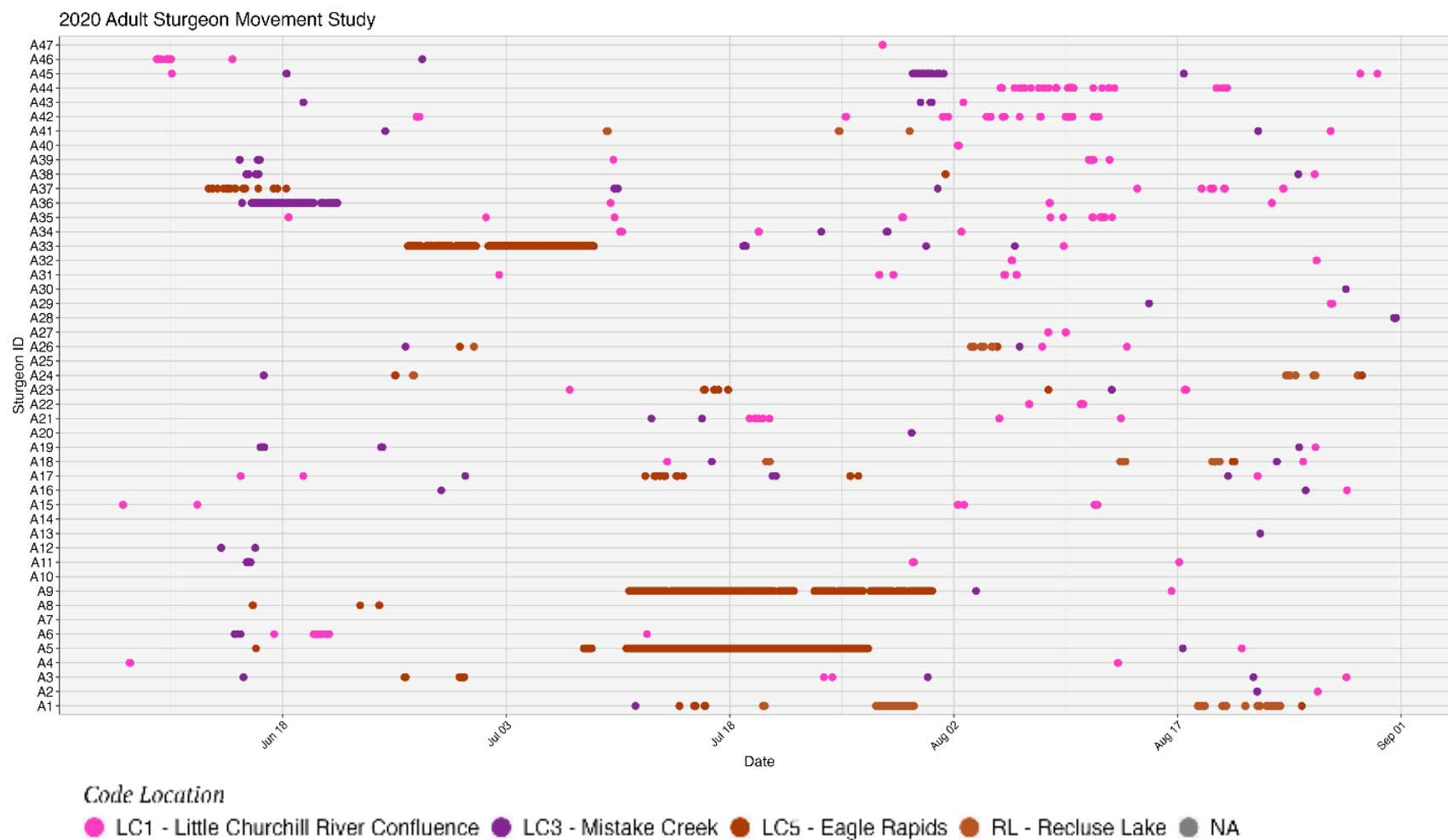


Figure A 8. Individual Lake Sturgeon movements tracked in 2020.

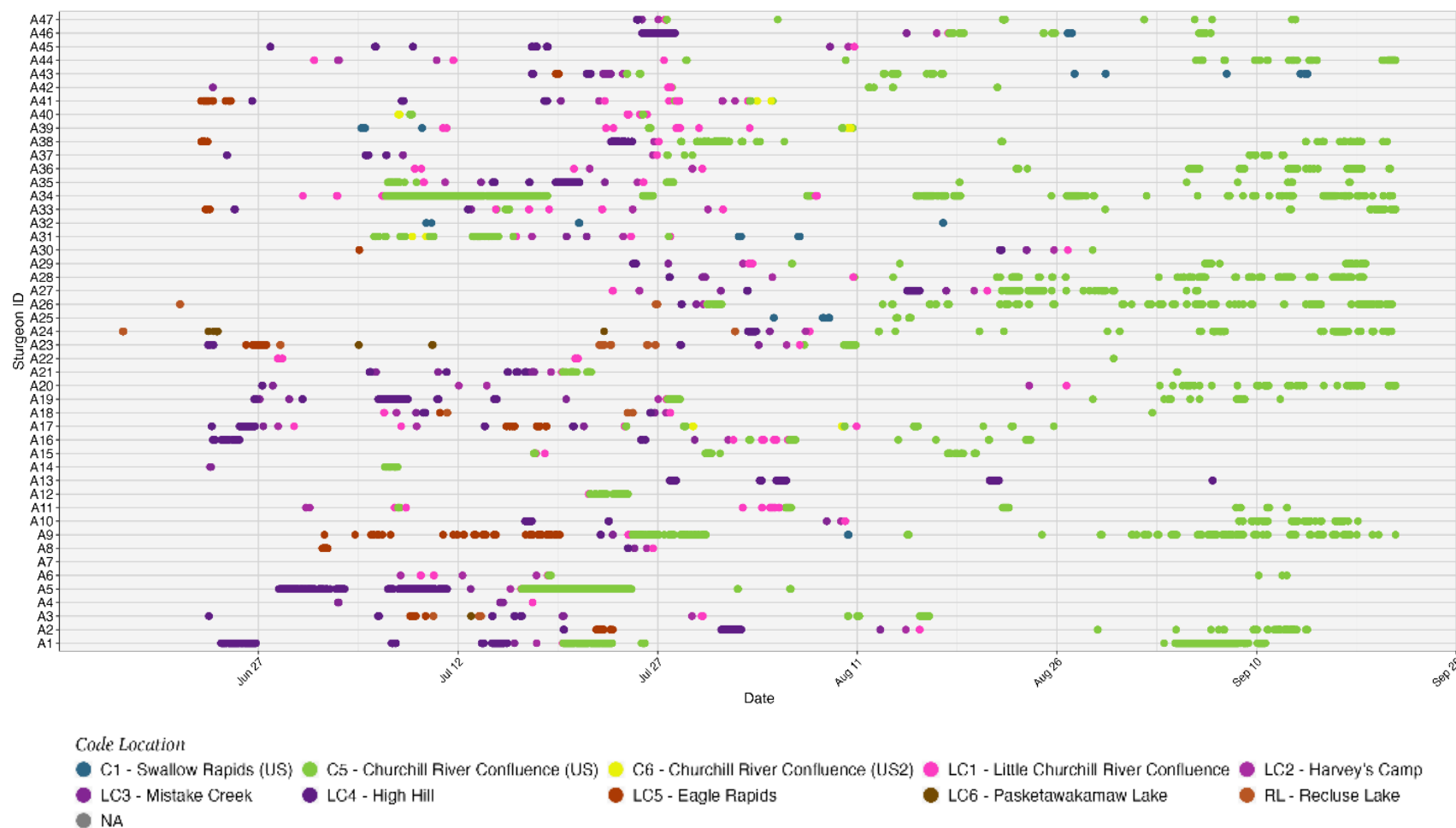


Figure A 9. Individual Lake Sturgeon movements tracked through 2021.

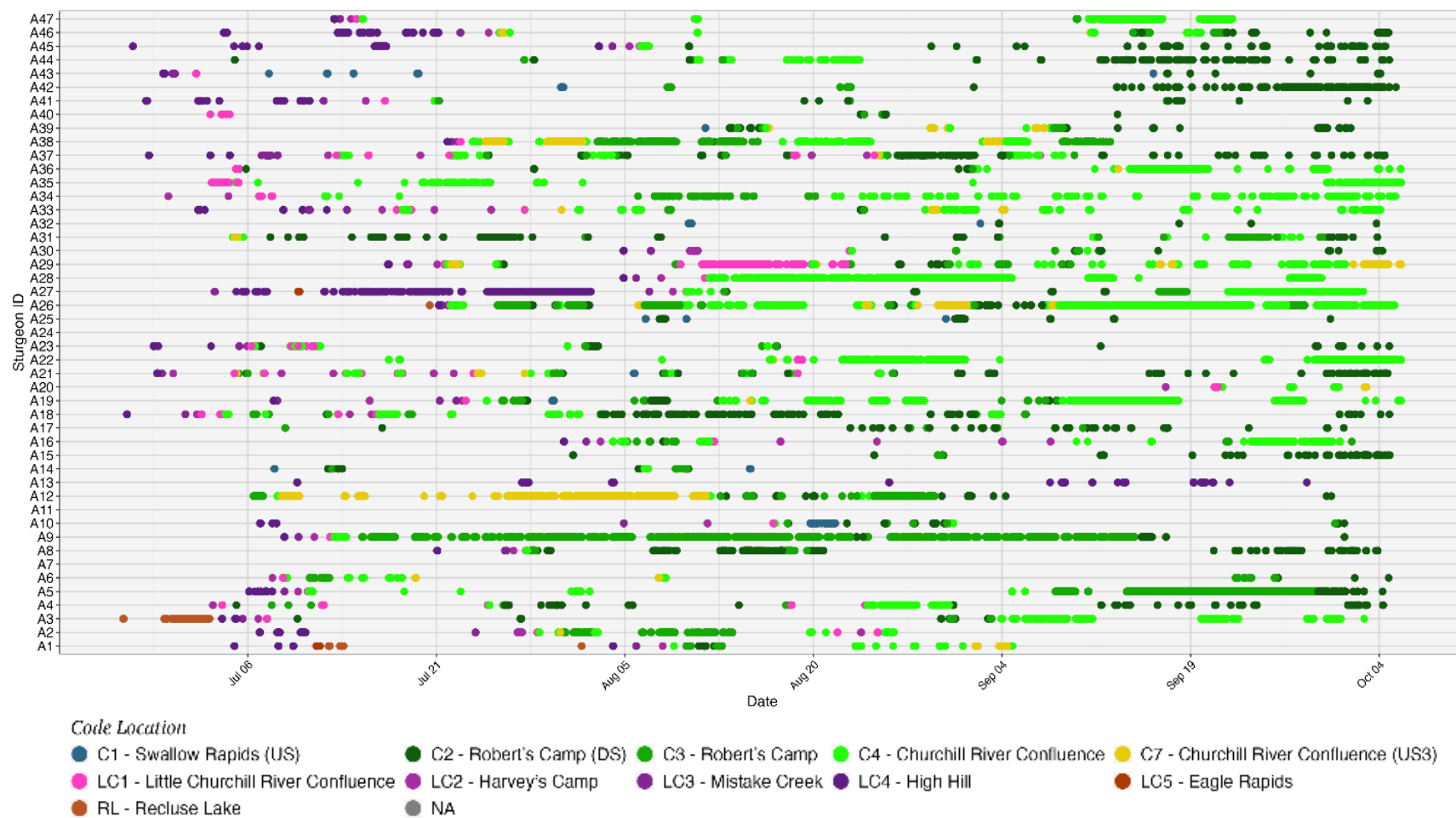


Figure A 10. Individual Lake Sturgeon movements tracked through 2022.

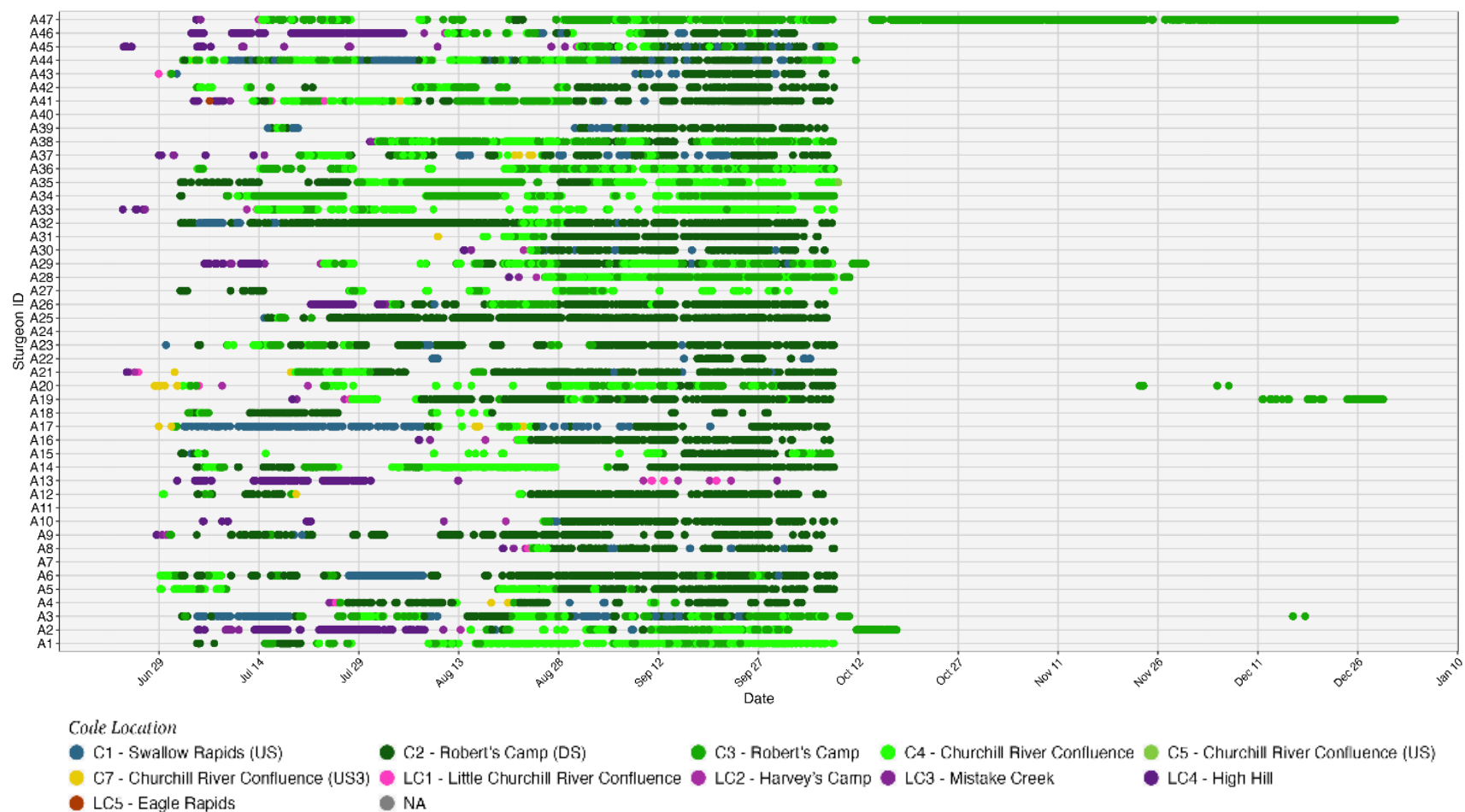


Figure A 11. Individual Lake Sturgeon movements tracked through 2023.

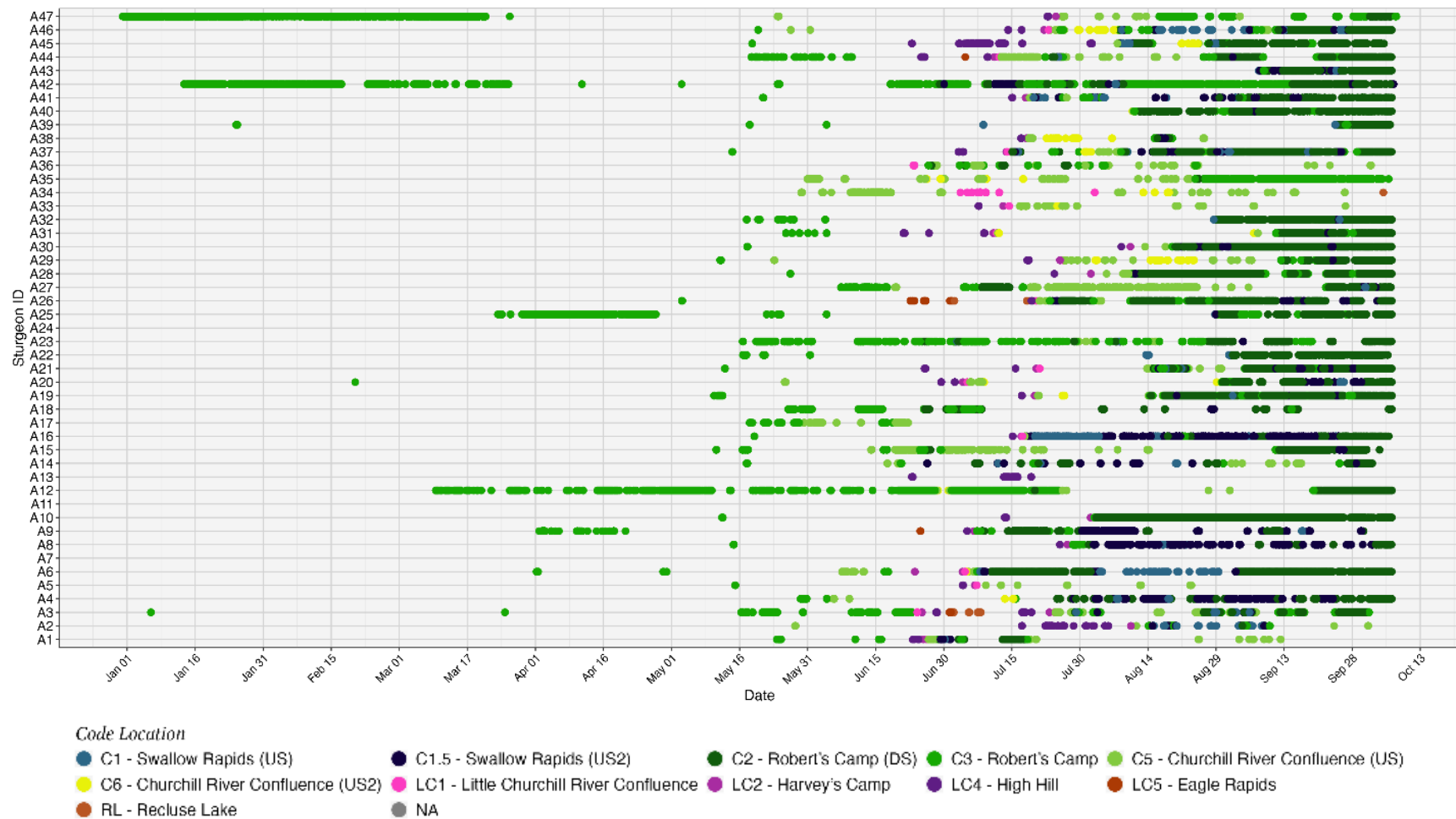


Figure A 12. Individual Lake Sturgeon movements tracked through 2024.

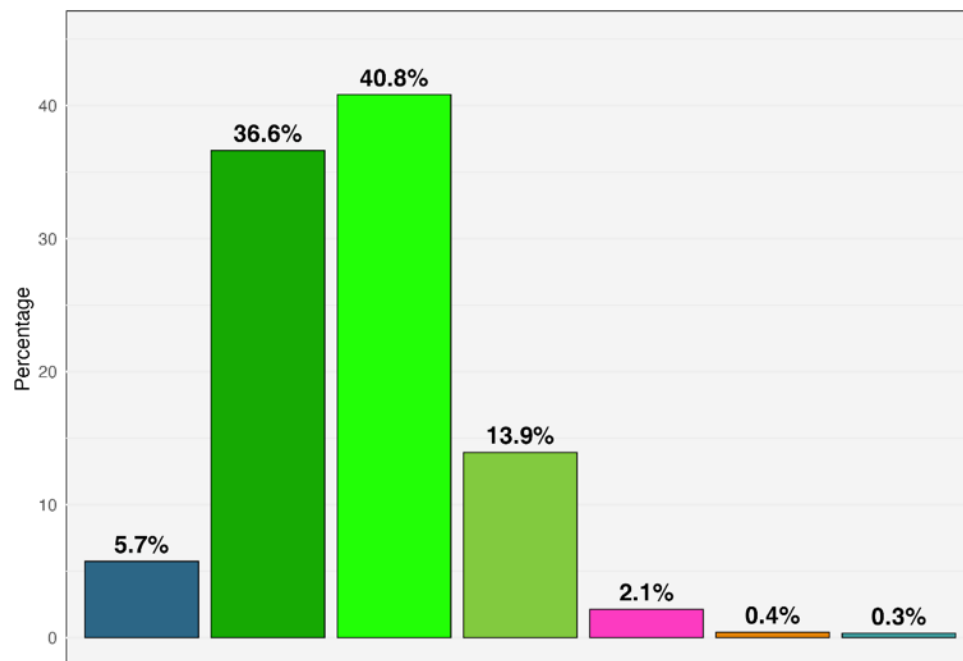


Figure A 13. Percentage of detections at each receiver during 2019. From left to right: C1 - Swallow Rapids, C3 - Robert's Camp, C4 - Churchill River Confluence, C5 - Churchill River Confluence (US), LC1 - Little Churchill River Confluence, MR - Mountain Rapids, SR-DS - Swallow Rapids (DS).

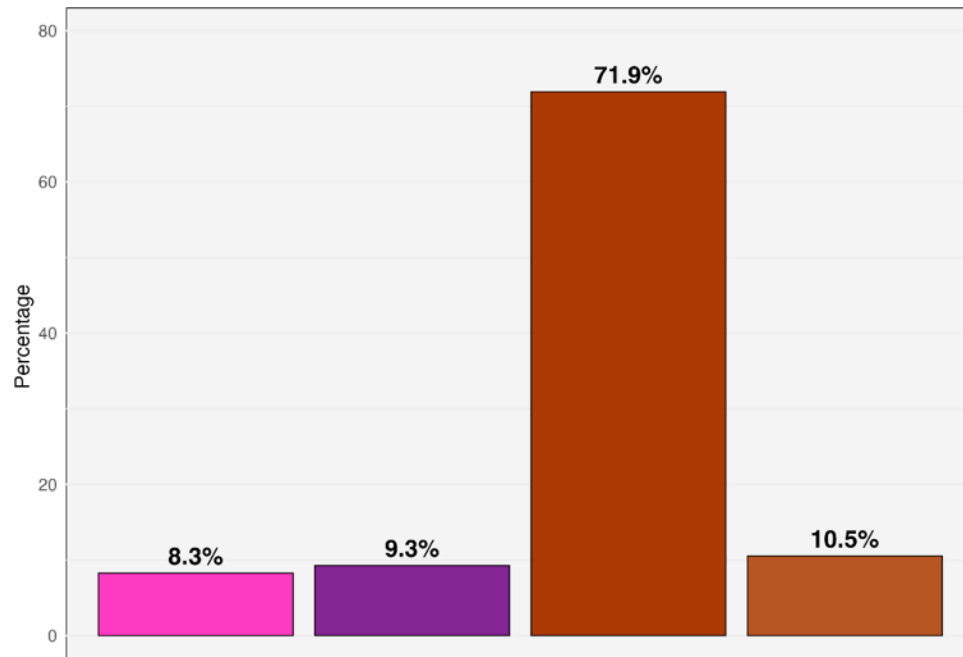


Figure A 14. Percentage of detections at each receiver during 2020. From left to right: LC1 - Little Churchill River Confluence, LC3 - Mistake Creek, LC5 - Eagle Rapids, RL - Recluse Lake.

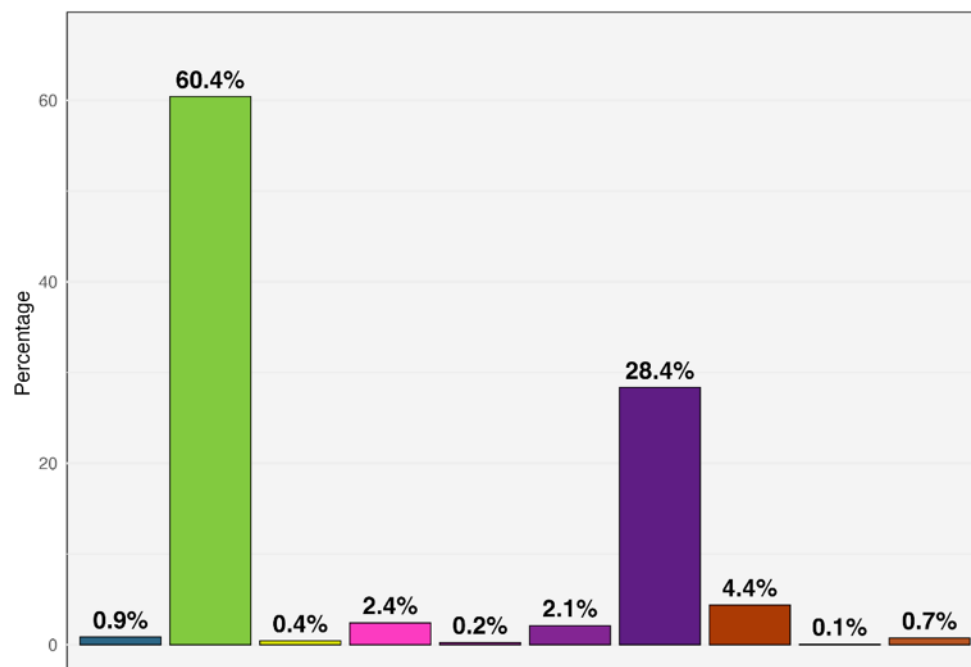


Figure A 15. Percentage of detections at each receiver during 2021. From left to right: C1 - Swallow Rapids, C5 - Churchill River Confluence (US), C6 - Churchill River Confluence (US2), LC1 - Little Churchill River Confluence, LC2 - Harvey's Camp, LC3 - Mistake Creek, LC4 - High Hill, LC5 - Eagle Rapids, LC6 - Pasketawakamaw Lake, RL - Recluse Lake.

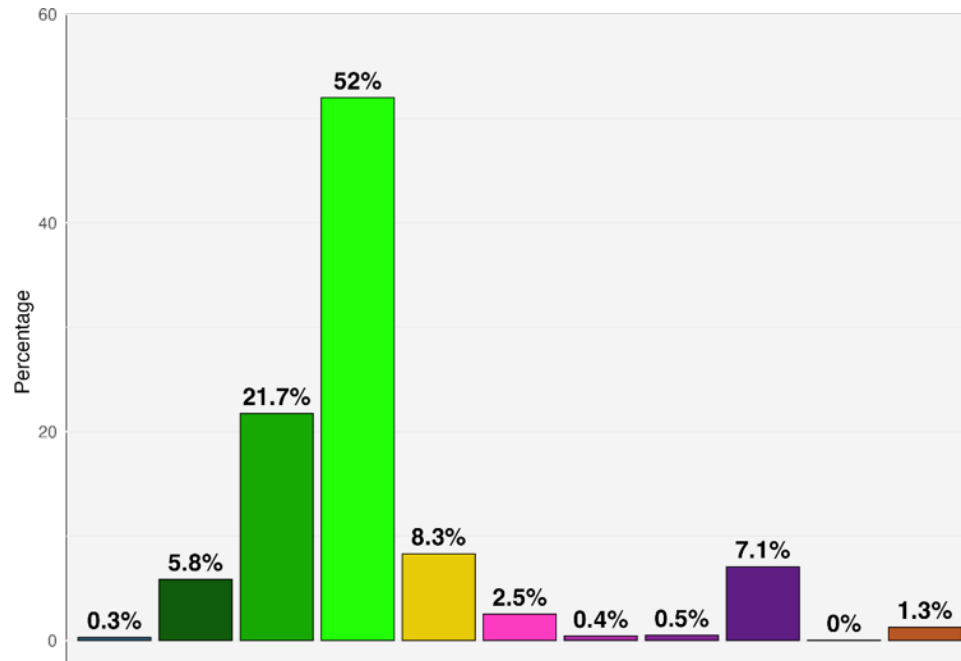


Figure A 16. Percentage of detections at each receiver during 2022. From left to right: C1 - Swallow Rapids, C2 - Robert's Camp (DS), C3 - Robert's Camp, C4 - Churchill River Confluence, C7 - Churchill River Confluence (US3), LC1 - Little Churchill River Confluence, LC2 - Harvey's Camp, LC3 - Mistake Creek, LC4 - High Hill, LC5 - Eagle Rapids, RL - Recluse Lake.

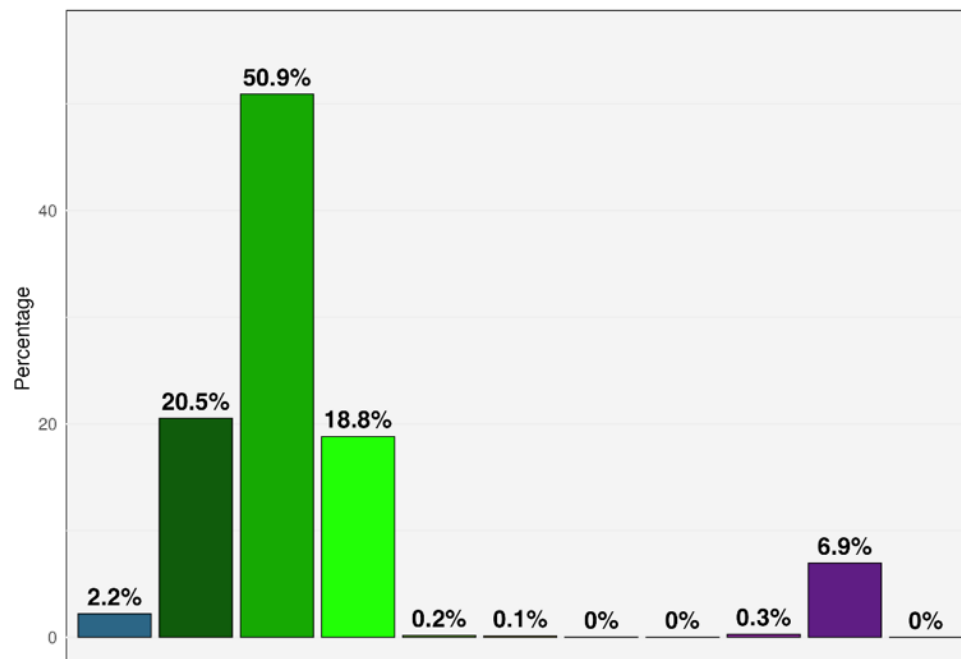


Figure A 17. Percentage of detections at each receiver during 2023. From left to right: C1 - Swallow Rapids, C2 - Robert's Camp (DS), C3 - Robert's Camp, C4 - Churchill River Confluence, C5 - Churchill River Confluence (US), C7 - Churchill River Confluence (US3), LC1 - Little Churchill River Confluence, LC2 - Harvey's Camp, LC3 - Mistake Creek, LC4 - High Hill, LC5 - Eagle Rapids.

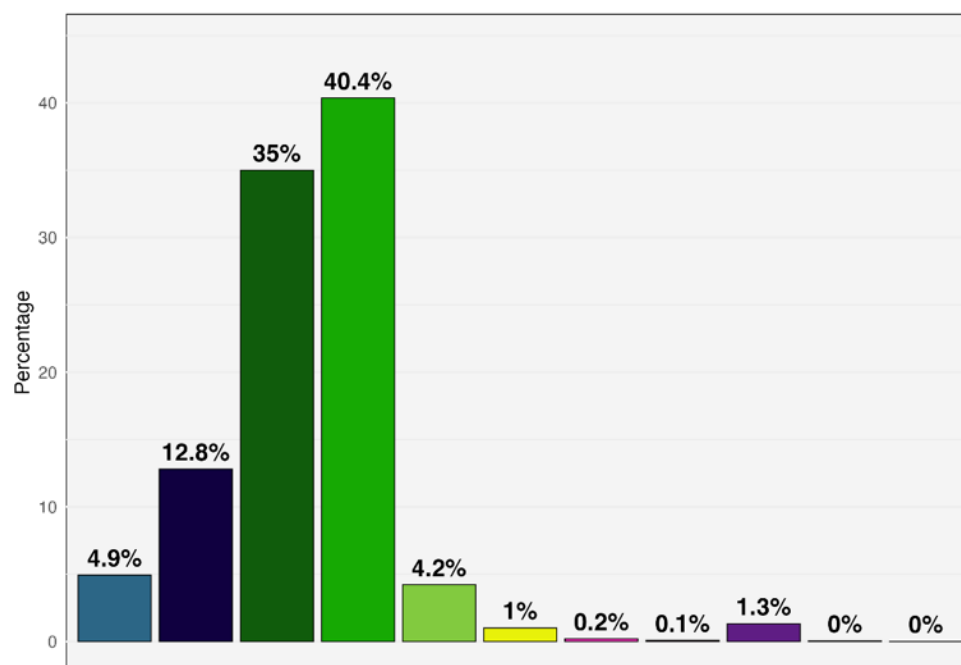


Figure A 18. Percentage of detections at each receiver during 2024. From left to right: C1 - Swallow Rapids, C1.5 - Swallow Rapids (US2), C2 - Robert's Camp (DS), C3 - Robert's Camp, C5 - Churchill River Confluence (US), C6 - Churchill River Confluence (US2), LC1 - Little Churchill River Confluence, LC2 - Harvey's Camp, LC4 - High Hill, LC5 - Eagle Rapids, RL - Recluse Lake.

Appendix A3 – Reference Photographs



Figure A 19. Vemco VR2Tx Acoustic Receiver anchored to a large stone prior to deployment for tracking Lake Sturgeon movement throughout the Lower Churchill River watershed.



Figure A 20. AAE biologist travelling the course of the Little Churchill River from Waskaiowaka Lake to the confluence to deploy Vemco VR2Tx Acoustic Receivers, in June of 2021.



Figure A 21. AAE biologist preparing to deploy a Vemco VR2Tx Acoustic Receiver at Mistake Creek, on the Little Churchill River in June of 2021.



Figure A 22. Little Churchill River Didson sonar and mounting apparatus installed facing into west channel of the Little Churchill River, approximately 1 km upstream of the Churchill River. A weir constructed using snow fencing is used to direct fish swimming upstream into the field of view of the sonar.

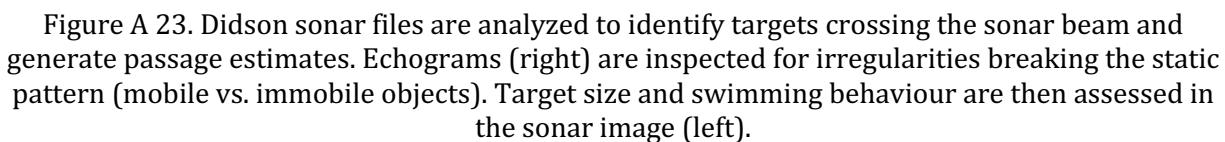




Figure A 25. Aerial view of low flow conditions at the Churchill – Little Churchill Confluence in the Fall of 2021.

Appendix A4 – Didson Sonar Data and Results

SONAR DATA PROCESSING

Data File Processing

The piSonar files were processed in Sound Metrics Didson V.5.26.26 software by scanning through echograms to identify irregularities in the recorded pattern that could be associated with fish movement through the sonar beam. Briefly, as static objects appear in the echogram as stable, horizontal lines (static through time), moving targets can be identified as irregular patterns breaking, or interfering with, the static, horizontal pattern (see Appendix B; Fig. B6). Moving targets were then visually assessed using the software's split-screen display to view a playback of the sonar image alongside the echogram. This allowed the user to visually distinguish fish from other moving targets, such as debris drifting downstream.

Visual assessments of swimming behaviour and the software's measure tool were used to approximate the size of targets and identify adult Lake Sturgeon moving through the sonar beam. While the larger adult sturgeon were easily identified by their size, certain confounding factors complicated the identification of smaller individuals. First, the lower resolution image combined with the size of the field of view (FOV; approx. 40 m) produced by the low-frequency setting makes the identification of species through assessments of morphological features difficult. Second, other large-bodied species, including Northern Pike (*Esox lucius*), Walleye (*Sander vitreus*), and Lake Whitefish (*Coregonus clupeaformis*), are common throughout the river system, and adult individuals routinely reach sizes over 0.5 m which is large enough to possibly be mistaken for a Lake Sturgeon. Finally, fish community species apportionment (% representation of each species) in the lower Churchill and Little Churchill River is not known. Thus, a statistical approach to species identification and apportionment for fish observed is not feasible.

To address these issues, and because the primary interest of this investigation was to evaluate utilization of the Little Churchill River by spawning adult sturgeon, a minimum size threshold of 1 m, as measured using the Didson software measure tool, was used to identify Lake Sturgeon in sonar records. This size threshold was selected as an identification criteria for the following reasons: (1) length-at-age and age-at-maturity data for Lake Sturgeon in the Churchill River, as well as other populations in Manitoba suggest that a large proportion of the spawning population will be >18 years in age and, as such, >1 m in total length (AAE Tech Services Inc.; unpub. Block, 2001; Haxton and Findlay, 2009); (2) besides sturgeon, only Northern Pike may reach sizes up to or exceeding 1 m in length in the Churchill River (based on species accounts detailed in Stewart and Watkinson, 2004); and (3) while Northern Pike may reach sizes up to or exceeding 1 m, pike of this size are almost exclusively female

(Malette and Morgan, 2005; Margenau et al., 1998; Priegel and Krohn, 1975; Raat, 1988; Žiliukienė and Žiliukas, 2012). At the time of these investigations (mid-June) and with water temperatures within the range observed, large female Northern Pike are expected to have been associating with shallow, still-water, densely-vegetated areas for spawning (Casselman, 1996; Clark, 1950; Craig, 2008; Holm et al., 2009; Larsson et al., 2015; Raat, 1988; Stewart and Watkinson, 2004). Further, no large Northern Pike (≥ 1 m) were captured in large-mesh gill nets set in the Little Churchill River in proximity to the Didson sonar. It is therefore likely that fish identified in sonar files measuring ≥ 1 m are mature adult Lake Sturgeon.

1.1.1 Didson Passage Estimates

Daily passage estimates for Lake Sturgeon migrating into the Little Churchill River during the spring spawning period were then calculated as the combined total of net upstream sturgeon counted from individual sonar files for each day of recording. Briefly, “net upstream” hourly counts were calculated as the difference between the number of sturgeon observed passing through the sonar beam with an upstream trajectory and those passing with a downstream trajectory. In cases where total file duration was less than 60 minutes as a result of sonar outages (e.g. generator fueling, maintenance), hourly counts were adjusted by correcting for file time, as follows:

$$N_h = \frac{C_u - C_d}{(t/60)}$$

Where:

N_h = hourly estimate

C_u = total upstream count

C_d = total downstream count

t = file duration (in minutes)

Where hourly counts were missing entirely due to outages >60 minutes, estimates were generated by calculating the average of net upstream counts from as many hours before and after the outage as the number of hours missing. In other words, if two hours of recording were missing, the estimate for each of these hours would be calculated as the average of net upstream counts from two hours before and two hours after the outage. Raw daily passage estimates were calculated as the sum of all hourly estimates for each day of recording:

$$N_d = \sum_{h=1}^n N_h$$

Where:

N_d = Daily passage estimate

N_h = Hourly passage estimate at hour h

n = Number of hourly estimates per day of recording

h = Individual hours of recording, from first (h_1) to last (h_n)

Finally, these daily estimates were corrected to account for differences between days of recording in the total amount time the sonar was operational, as follows:

$$D_n = \frac{N_d}{t_d/1440}$$

Where:

D_n = Corrected daily passage estimate

N_d = Daily passage estimate on day d

t_d = total minutes of recording on day d

Correlations between sturgeon passage into the Little Churchill River and water temperature in the Little Churchill River (Little Churchill LC1 Vemco transceiver), and between sturgeon passage, discharge and primary water level in both the Churchill and Little Churchill rivers (WSC gauging stations 06FB001 - Churchill River below Fidler Lake & 06FC001 - Little Churchill River below Recluse Lake) were assessed by Pearson's correlation analysis. When data did not meet the assumption of normality (Shapiro-Wilk normality test and visual assessment of Q-Q plots), Spearman's non-parametric analysis of correlation was used instead. A value of $p < 0.05$ was used as a threshold for significance in all tests.

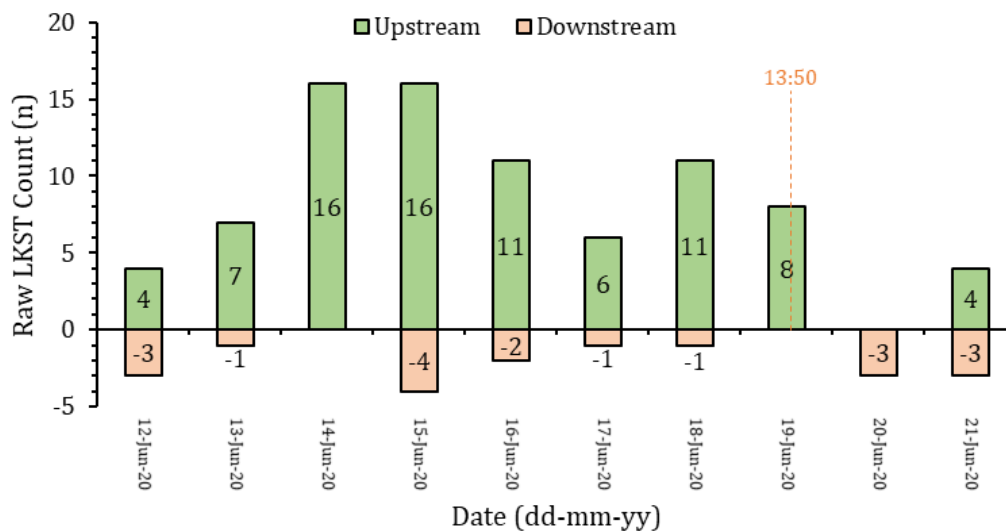


Figure A 26. Raw upstream (+) and downstream (-) counts of adult Lake Sturgeon identified in sonar records from the Little Churchill River between June 12th and June 21st, 2020. The dashed line indicates the time when the sonar was relocated to the east channel.

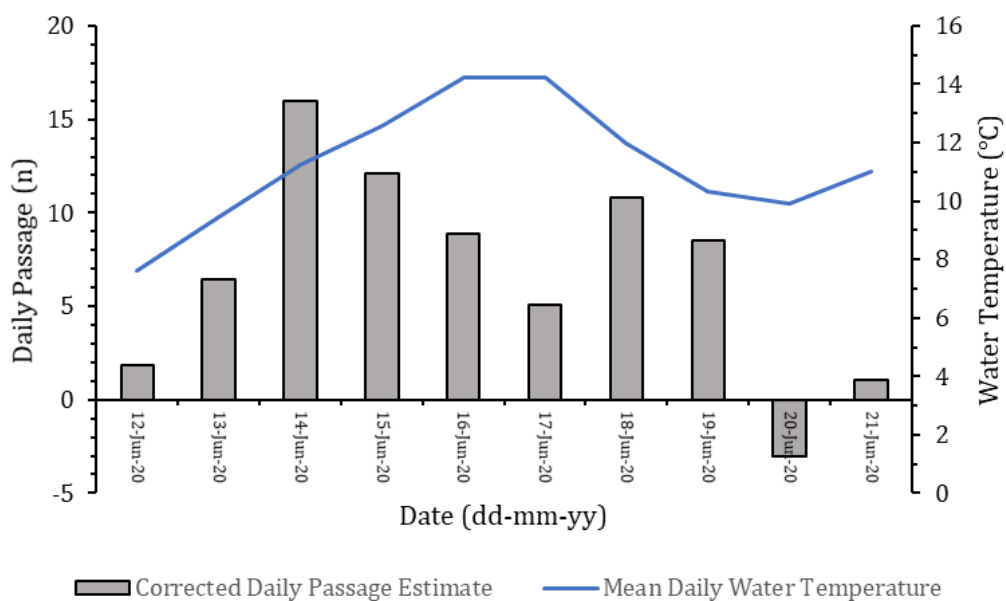


Figure A 27. Corrected daily passage estimates for adult Lake Sturgeon migrating into the Little Churchill River between June 12th and June 21st, 2020. Average daily water temperatures are calculated from hourly logs recorded by the LC1 Vemco transceiver. The sonar was relocated to the east channel on June 19, 2020.

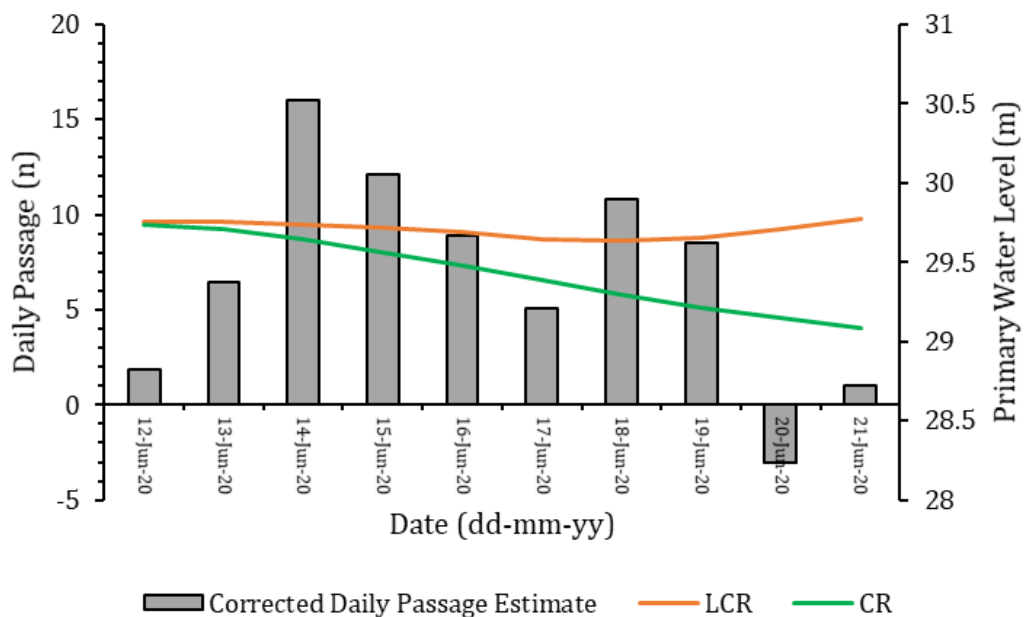


Figure A 28. Corrected daily passage estimates for adult Lake Sturgeon migrating into the Little Churchill River between June 12th and June 21st, 2020. Primary water levels in the Little Churchill River (LCR; orange) and Churchill River (CR; Green) are presented as daily averages calculated from real-time hydrometric data obtained from WSC gauging stations 06FB001 and 06FC001 (Government of Canada, accessed 2021).

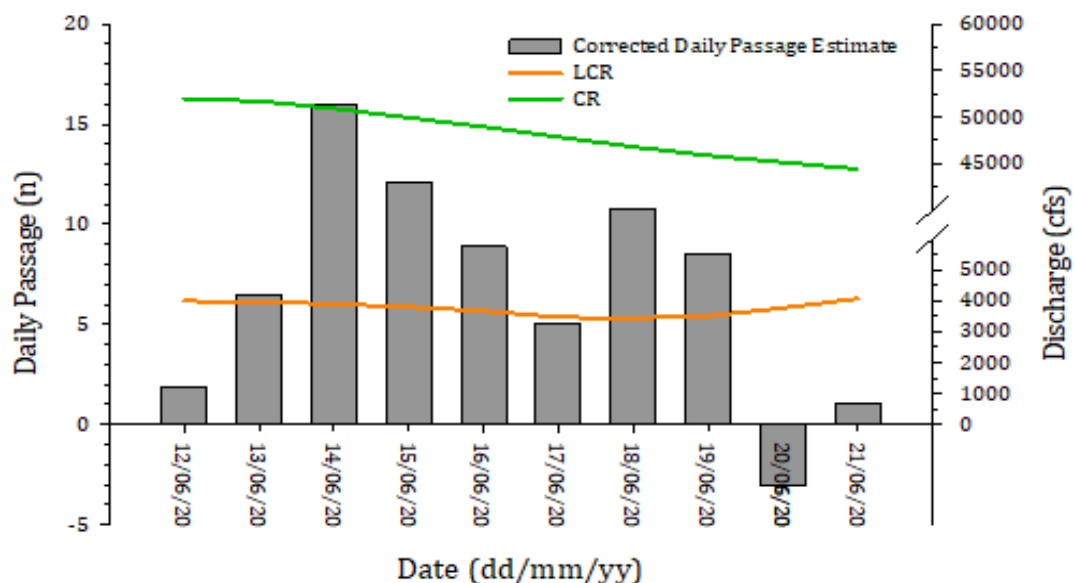


Figure A 29. Corrected daily passage estimates for adult Lake Sturgeon migrating into the Little Churchill River between June 12th and June 21st, 2020. Discharge in the Little Churchill River (LCR; orange) and Churchill River (CR; Green) are presented as daily averages calculated from real-time hydrometric data obtained from WSC gauging stations 06FB001 and 06FC001 (Government of Canada, accessed 2021).